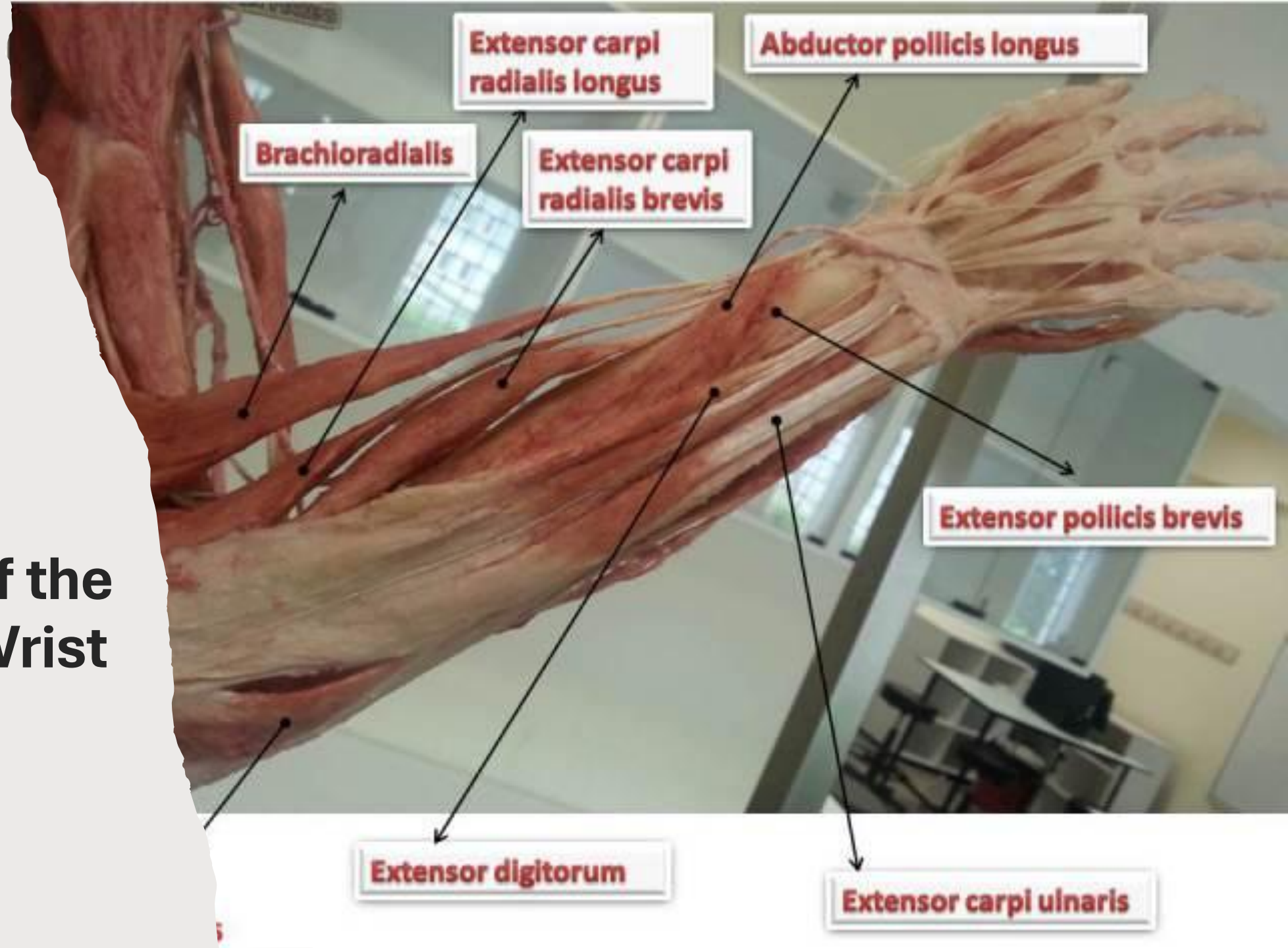
Anatomy of the Forearm, Wrist and Hand

Walaa Abu-Taleb



Learning Outcomes

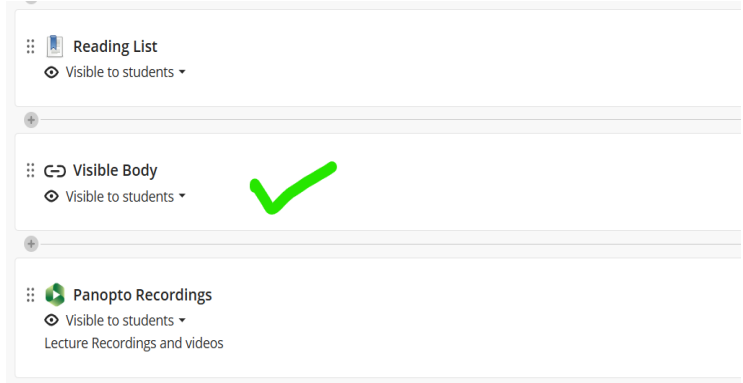
By the end of this session, students should be able to:

1. Identify the major muscles of the forearm, wrist and hand.

2. Describe the innervation and vascular supply of these muscles, correlating them with their respective nerves and arteries.

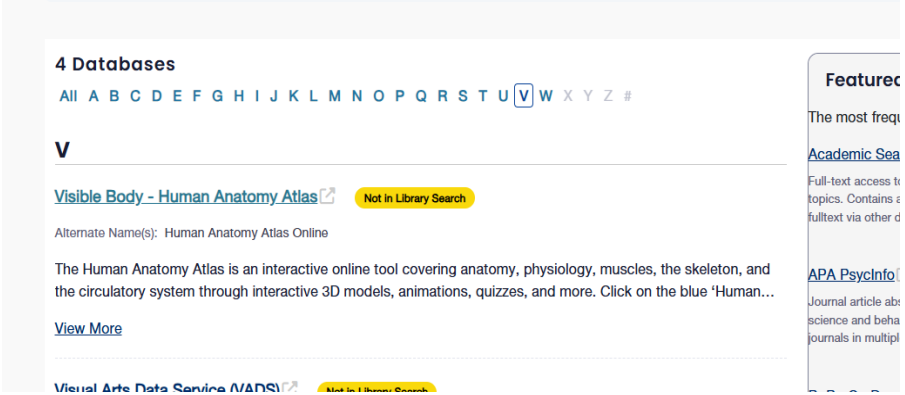
3. Apply knowledge of muscle function and neurovascular supply to clinical assessments, injury evaluation, and rehabilitation strategies.

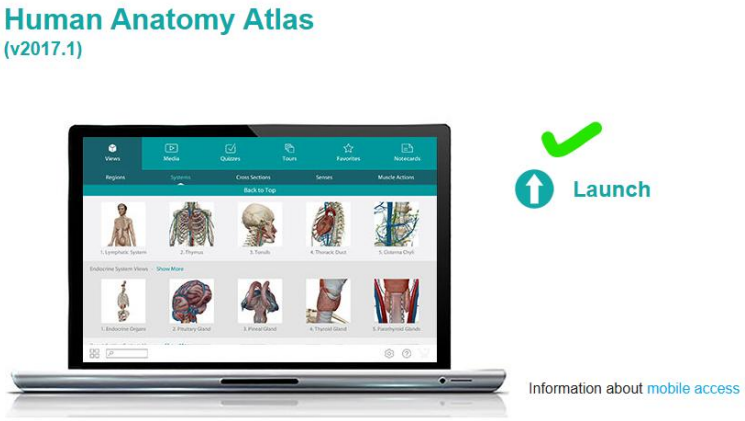
Practical Instructions



- In your module, click on the Visible Body Link

- Launch the *Visible Body – Human Anatomy Atlas*





Skeletal System Views ➔ Show More ➔ Shoulder Girdle

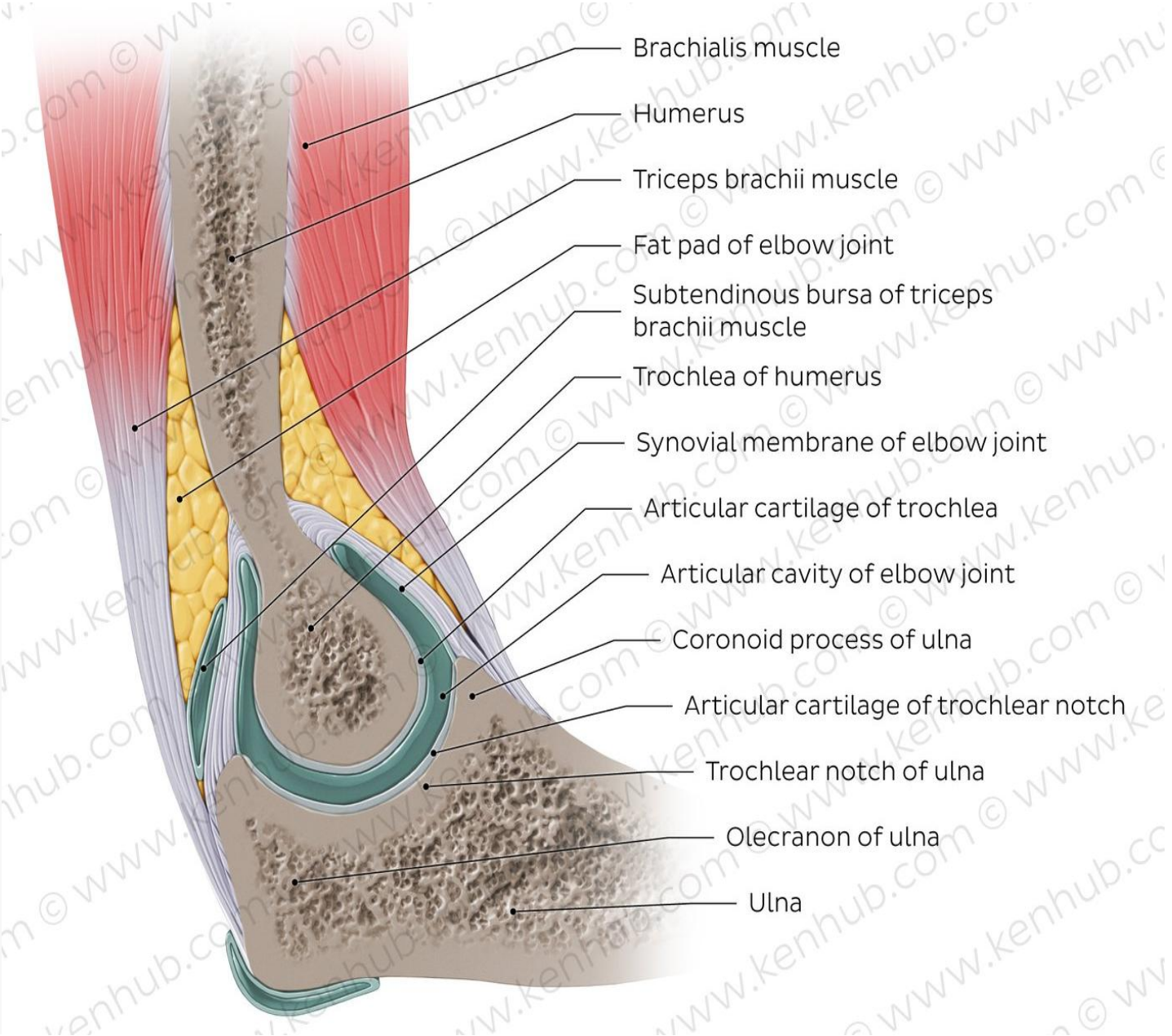
In your groups

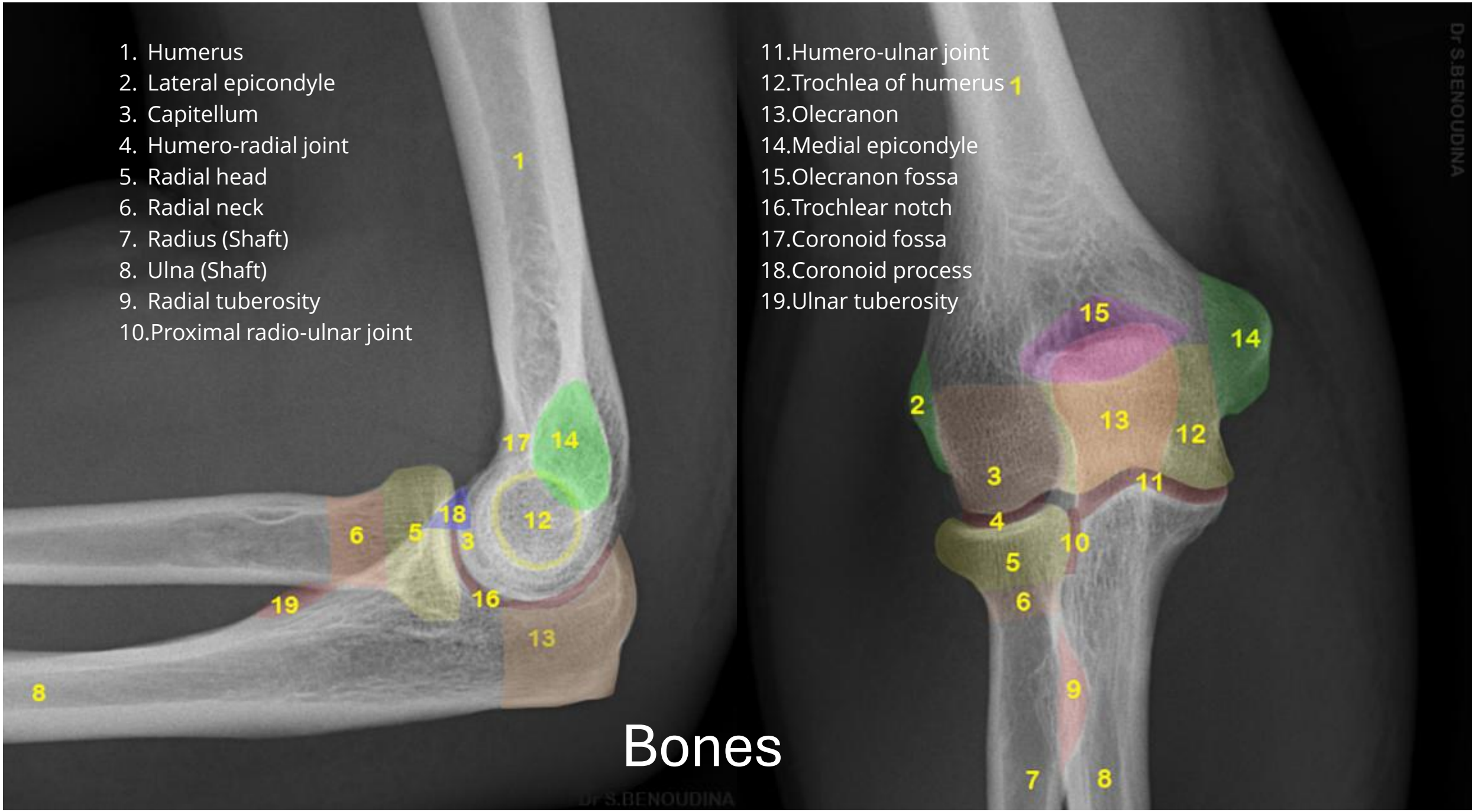
1. Identify the bones of the elbow joint and the forearm
2. Make use of the select tool and interact with the visible body system (3-D interaction)
3. Identify the ligaments that hold the bones together
4. Compare the images and match them to what you learned in Anatomy of the lecture earlier today.

Elbow Joint

The elbow joint consists of three joints; the humeroulnar joint, the humeroradial joint, and the proximal radioulnar joint, all within one articular capsule.

- Bones: humerus, radius, ulna
- Type: synovial, hinged joint
- Concave and convex sides
- Movements: 150 degrees
  - Flexion - biceps brachii, brachialis, brachioradialis muscles
  - Extension - triceps brachii muscle

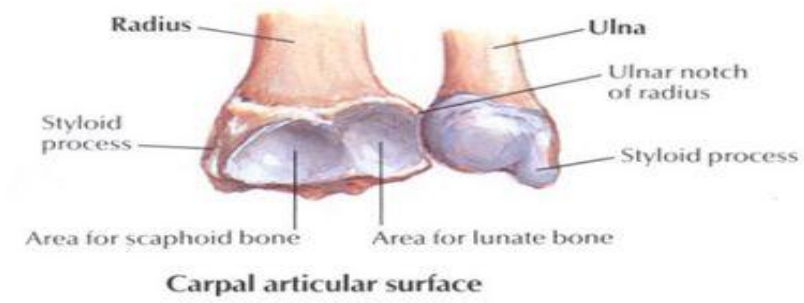


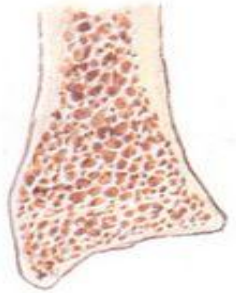

1. Humerus
2. Lateral epicondyle
3. Capitellum
4. Humero-radial joint
5. Radial head
6. Radial neck
7. Radius (Shaft)
8. Ulna (Shaft)
9. Radial tuberosity
10.Proximal radio-ulnar joint
11.Humero-ulnar joint
12.Trochlea of humerus
13.Olecranon
14.Medial epicondyle
15.Olecranon fossa
16.Trochlear notch
17.Coronoid fossa
18.Coronoid process
19.Ulnar tuberosity
Bones
Dr S.BENOUDINA

Bones







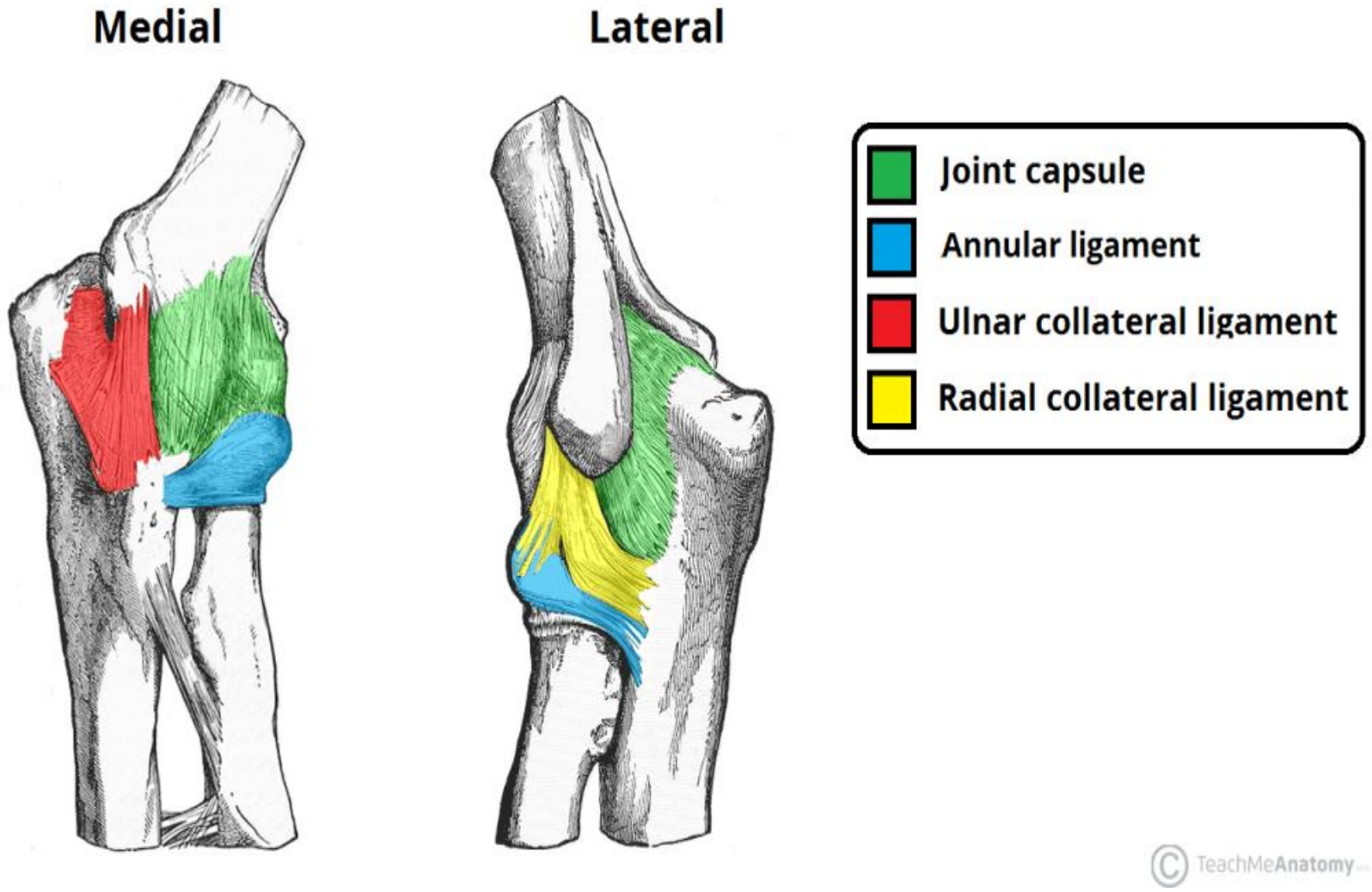


Coronal section of radius demonstrates how thickness of cortical bone of shaft diminishes to thin layer over cancellous bone at distal end

Ligaments

The elbow joint is supported by three ligaments:

- The annular ligament
- The radial collateral ligament
- The ulnar collateral ligament



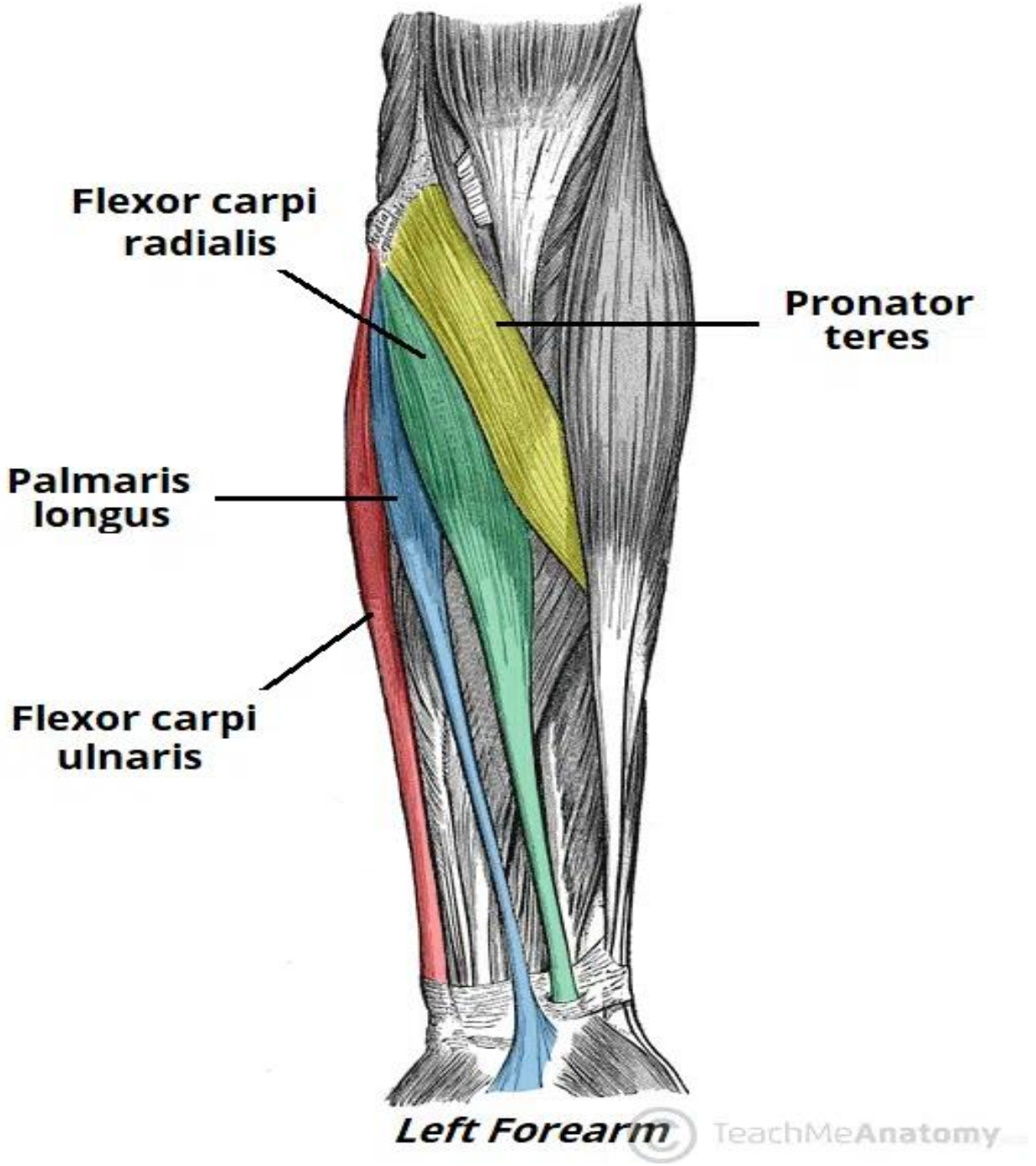
Anterior Forearm Muscles

Superficial: flexor carpi ulnaris, palmaris longus, flexor carpi radialis, pronator teres.

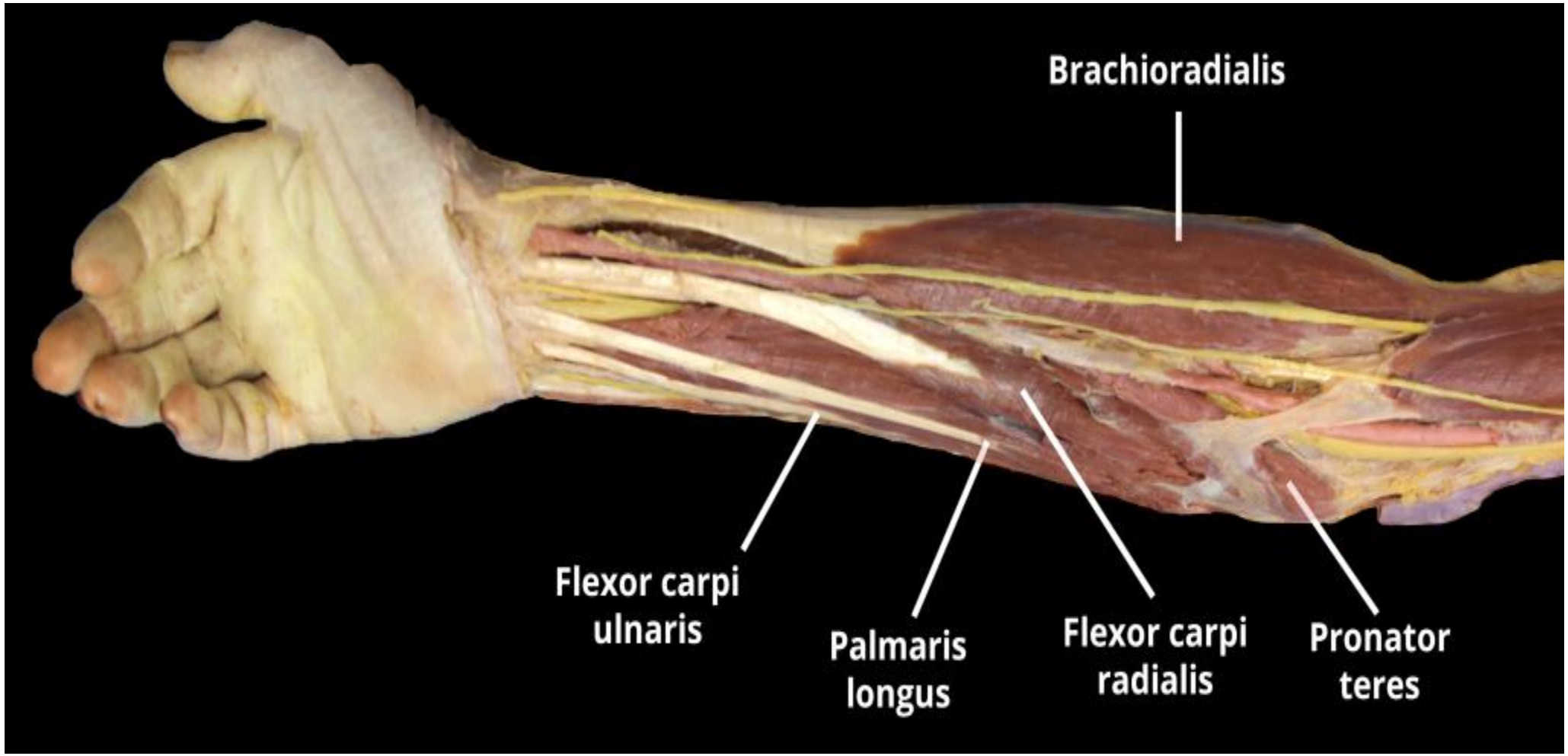
Intermediate: flexor digitorum superficialis.

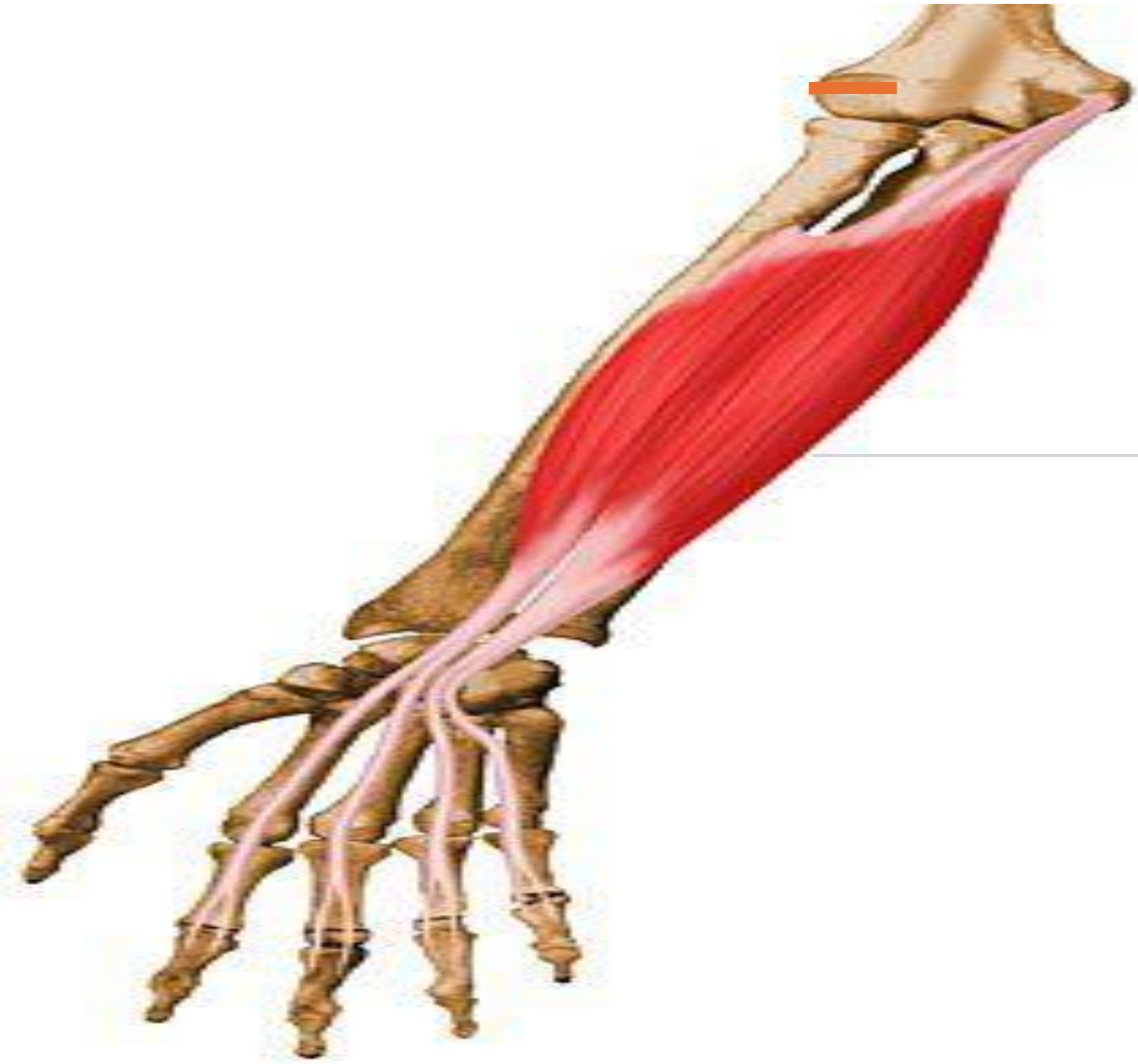
Deep: flexor pollicis longus, flexor digitorum profundus and pronator quadratus.

Superficial group



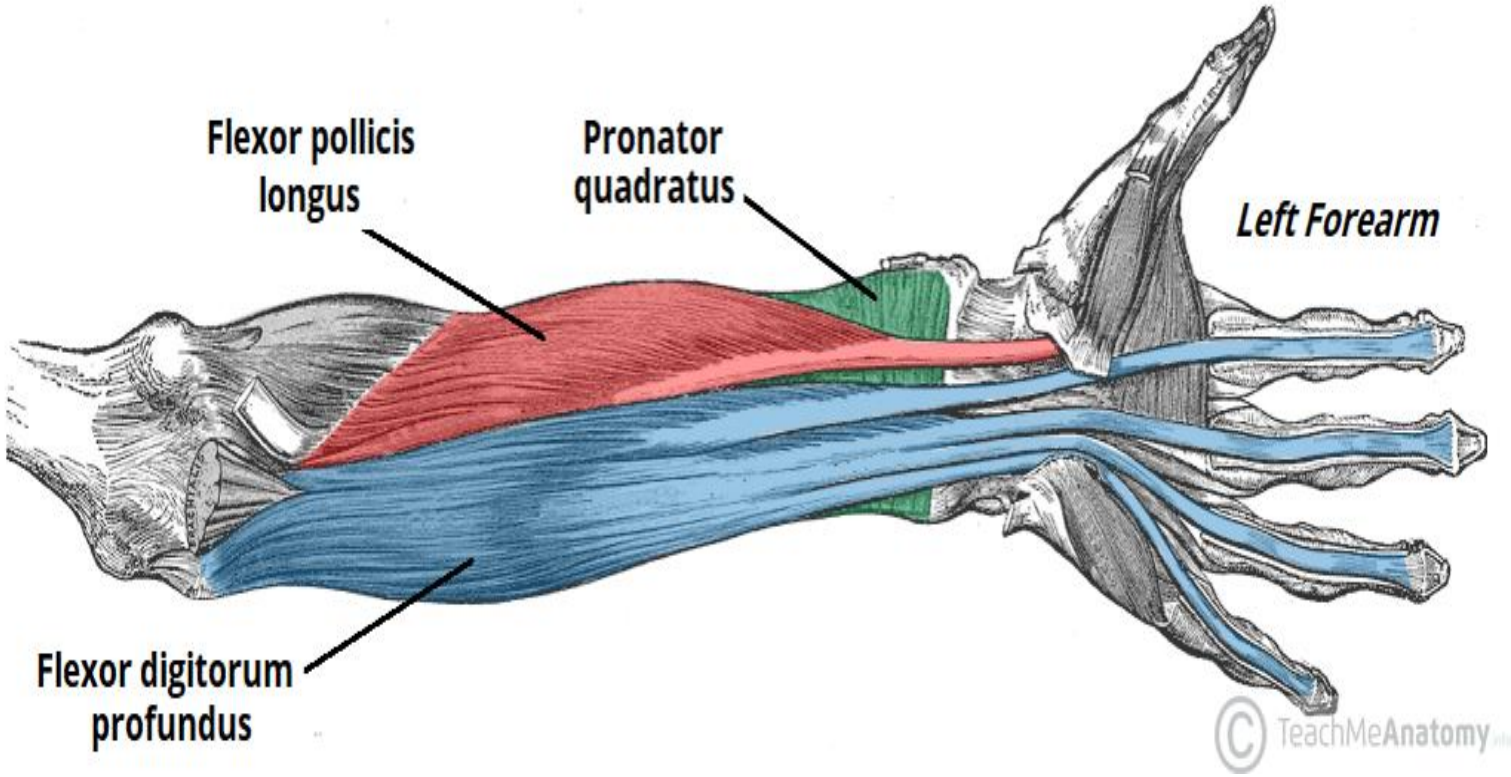
Dissection anatomy





Flexor Digitorum Superficialis

Deep Compartment



Explore the anterior forearm Compartment sin visible body app (10 minutes)

Superficial: flexor carpi ulnaris, palmaris longus, flexor carpi radialis, pronator teres.

Intermediate: flexor digitorum superficialis.

Deep: flexor pollicis longus, flexor digitorum profundus and pronator quadratus.

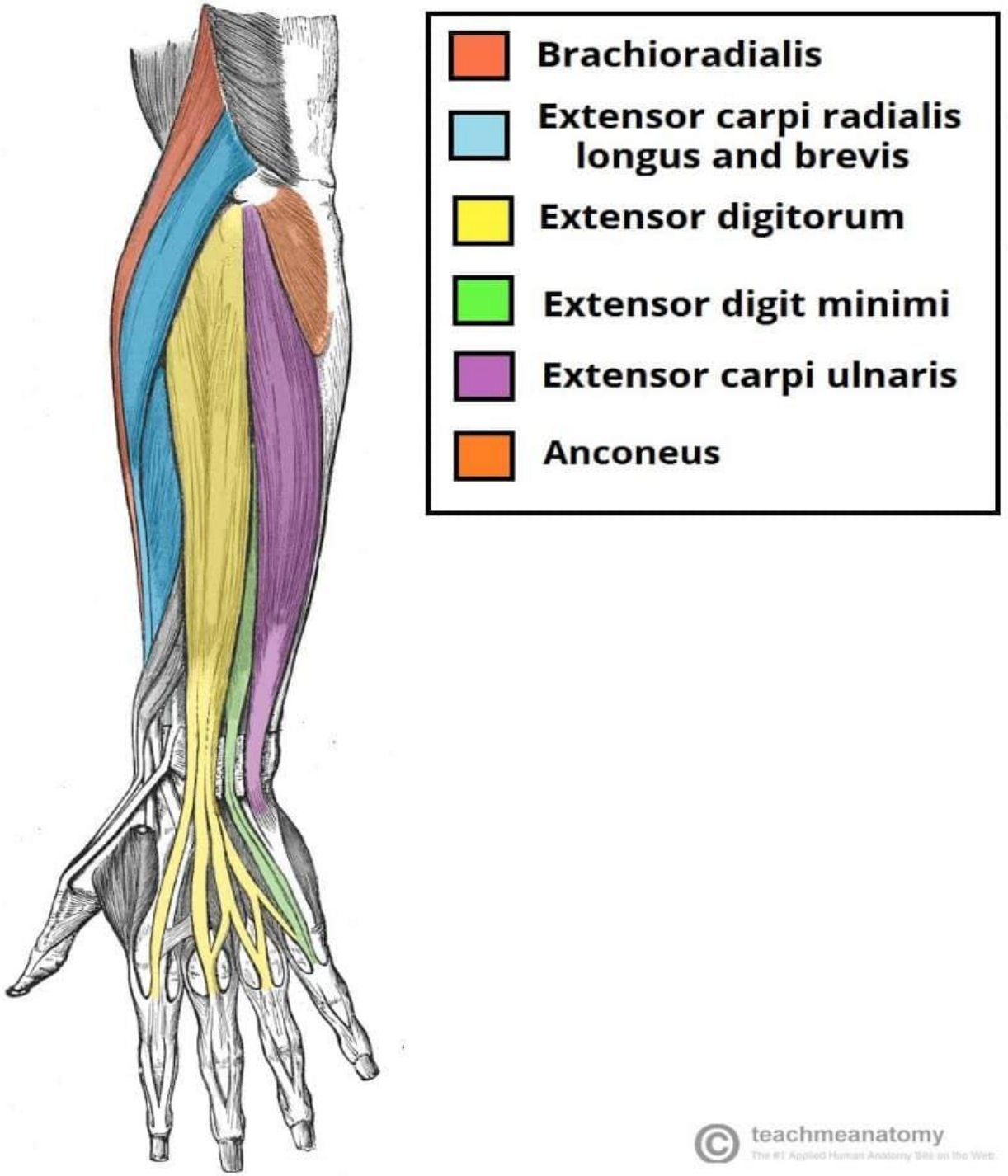


Extensors of the Forearm

(Dorsal Group)

Superficial Muscles

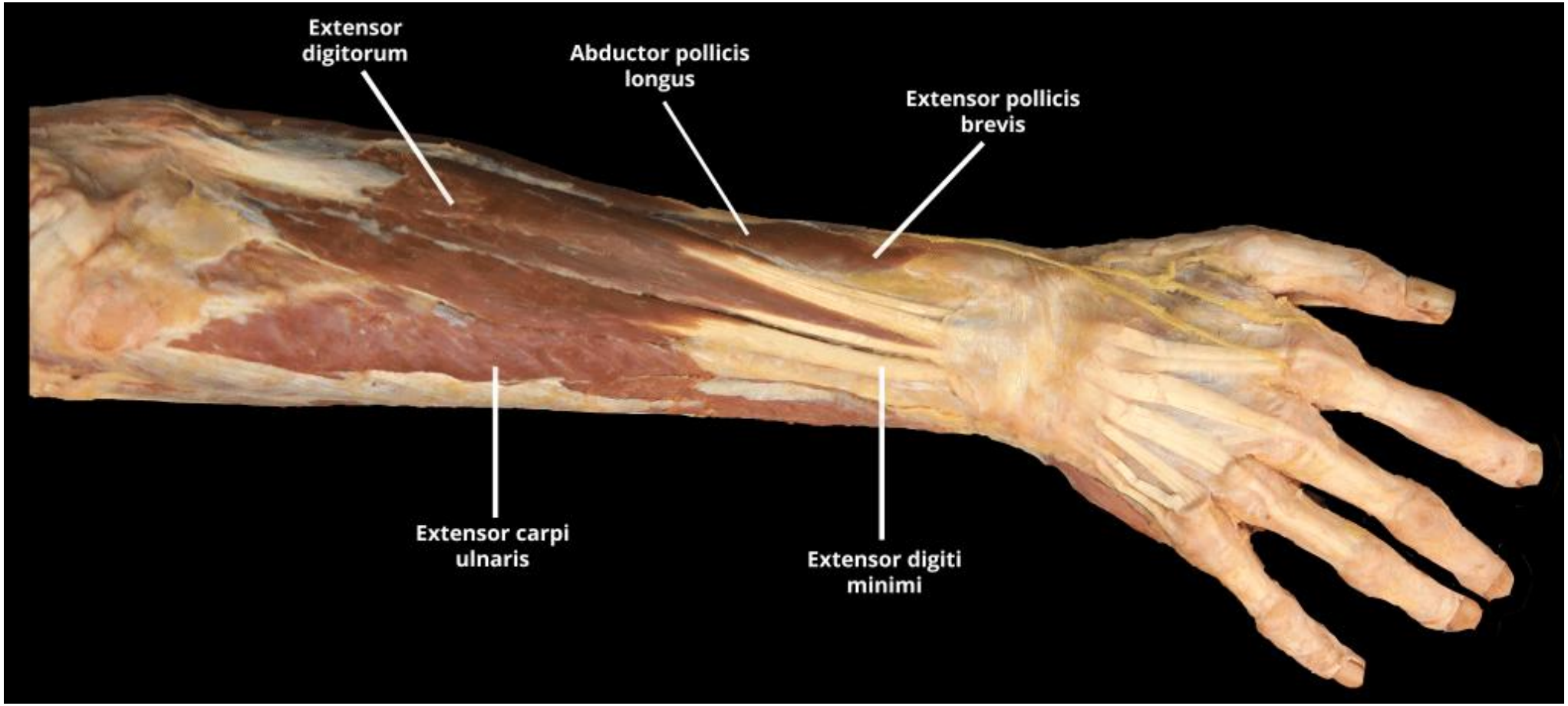
- The **superficial layer** of the posterior forearm contains seven muscles.



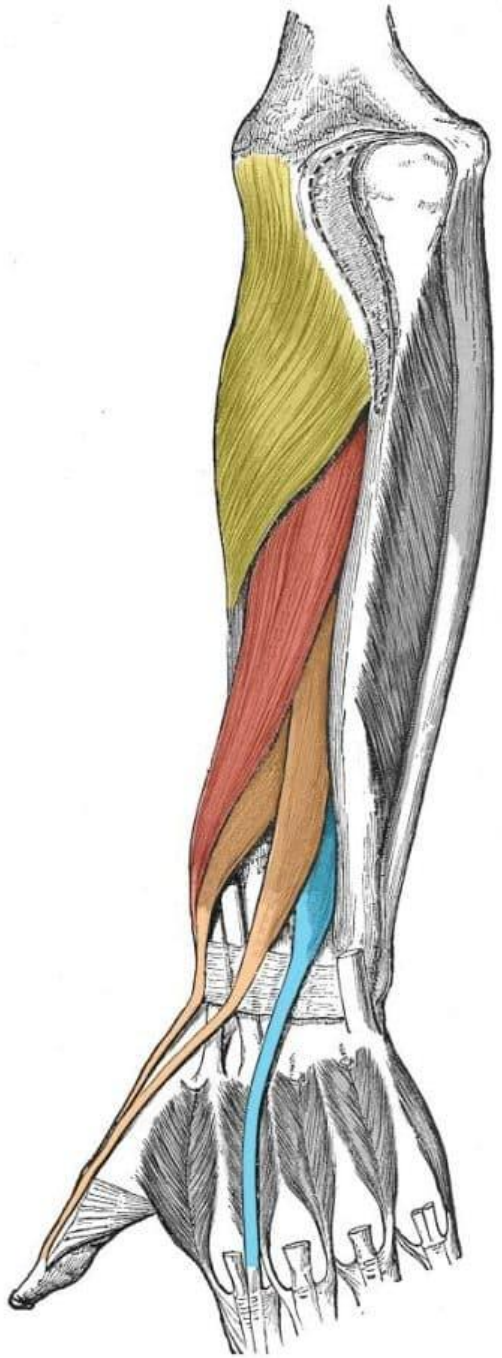
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- Brachioradialis
- Extensor carpi radialis longus and brevis
- Extensor digitorum
- Extensor digiti minimi
- Extensor carpi ulnaris
- Abductor pollicis longus
- Extensor pollicis longus


Extensor digitorum
Abductor pollicis longus
Extensor pollicis brevis
Extensor carpi ulnaris
Extensor digiti minimi

Deep Muscles

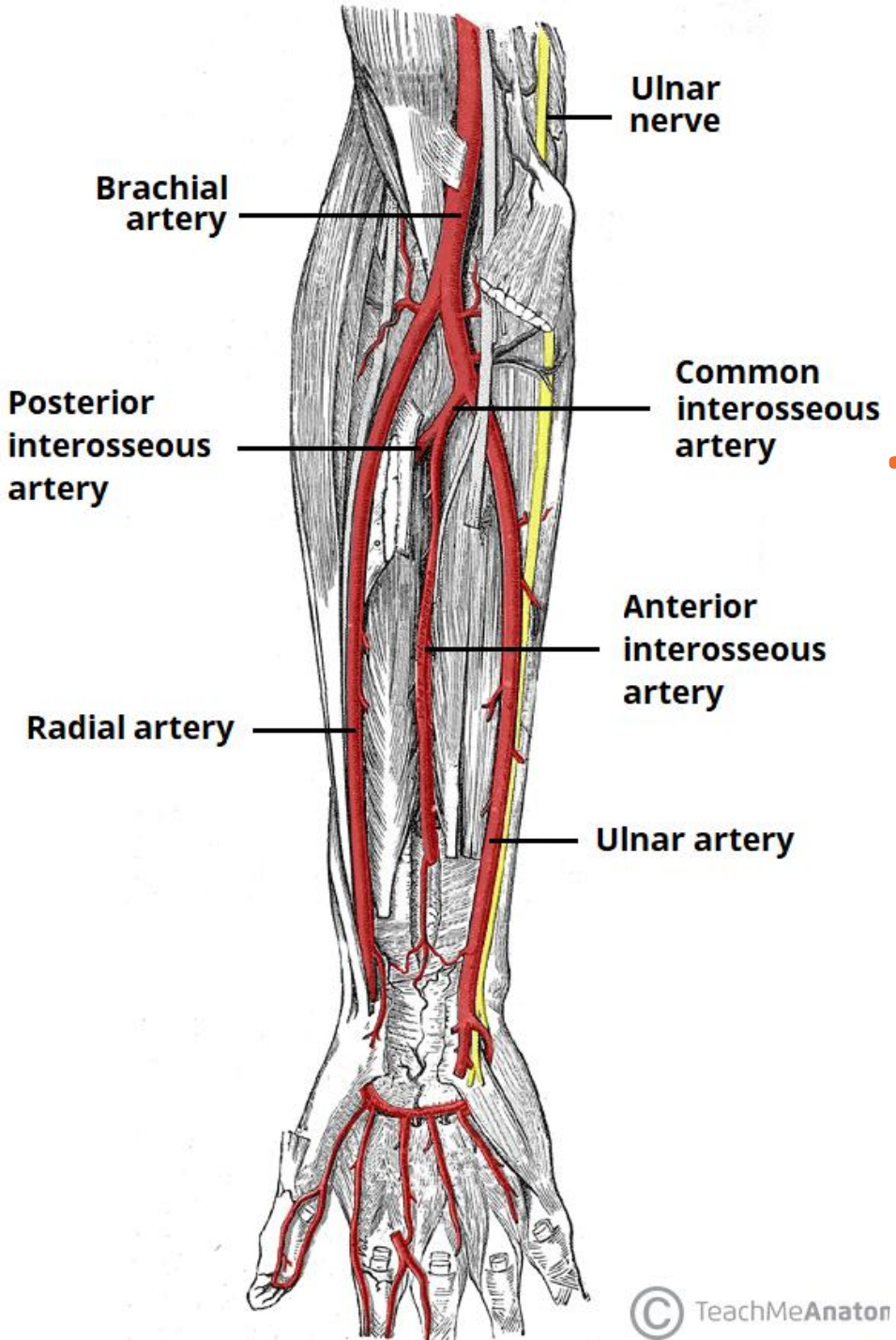


- Supinator
- Abductor pollicis longus
- Extensor pollicis longus and brevis
- Extensor indicis

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The #1 Applied Human Anatomy Site on the Web.

In your groups:

1. Identify the bones of the elbow joint and the forearm
2. Make use of the select tool and interact with the visible body system (3-D interaction)
3. Identify the ligaments that hold the bones together
4. Compare the images and match them to what you learned in Anatomy of the lecture earlier today.

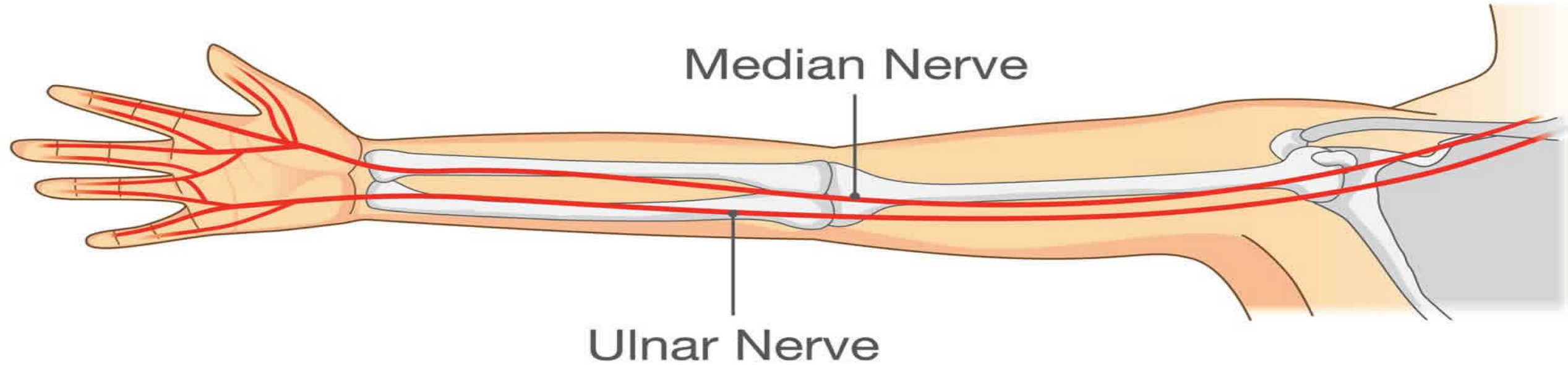


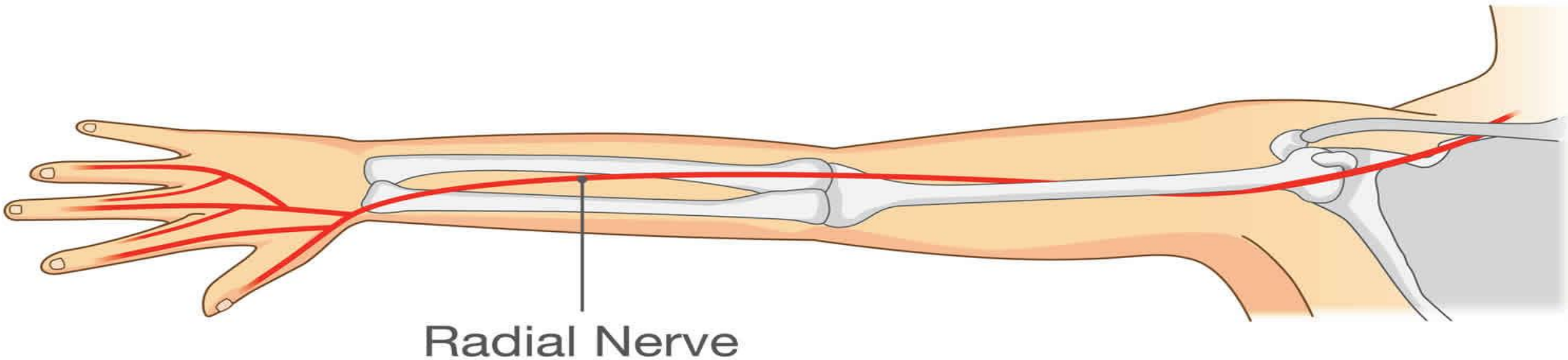
Radial and Ulnar Arteries

The radial and ulnar arteries are formed by the bifurcation of the brachial artery within the cubital fossa:

- Radial artery
- Ulnar artery
- Also gives rise to the anterior and posterior interosseous arteries
- These two arteries anastomose in the hand by forming two arches – the superficial palmar arch, and the deep palmar arch.

Nerve of the Arm





In your groups:

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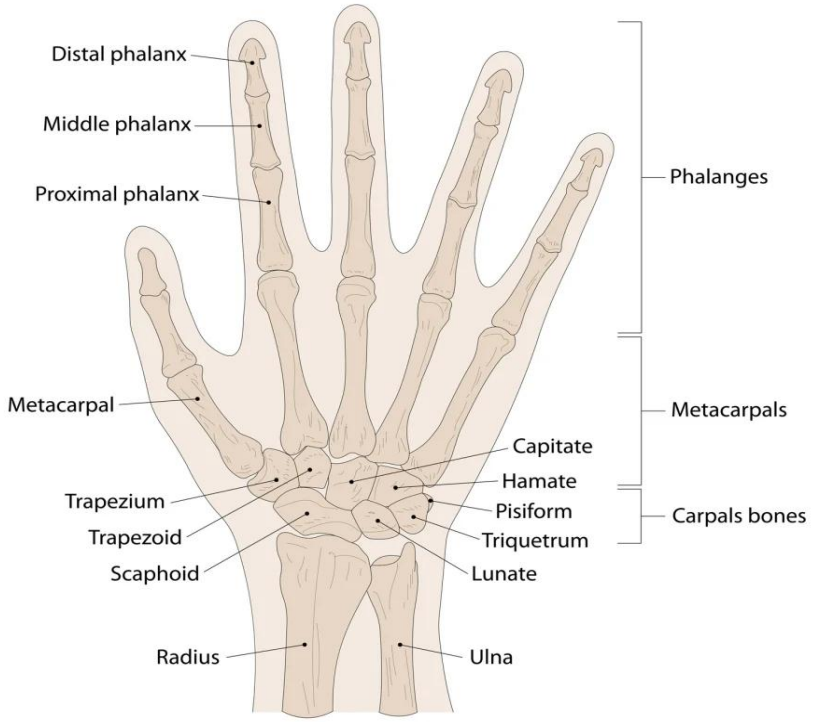


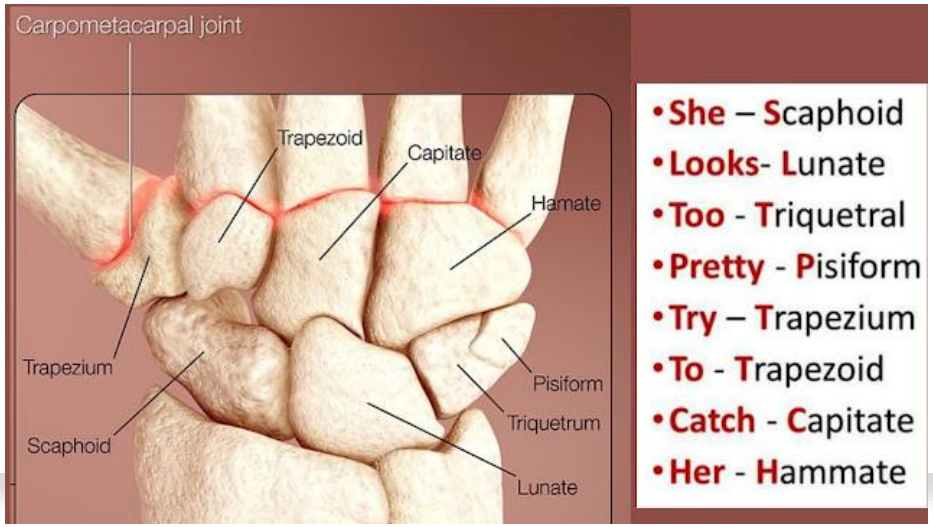
Hand Anatomy

Wrist Joint

There are 19 bones in each of the hands.

- Metacarpals: The bones that are in the palm and give it its shape.
- Phalanges: The individual bones that make up the segments of the fingers and thumb.
- Sesamoids: Small bones embedded in the tendons that help them move smoothly.
- the wrist is a complex joint made of eight bones that are arranged into two rows.

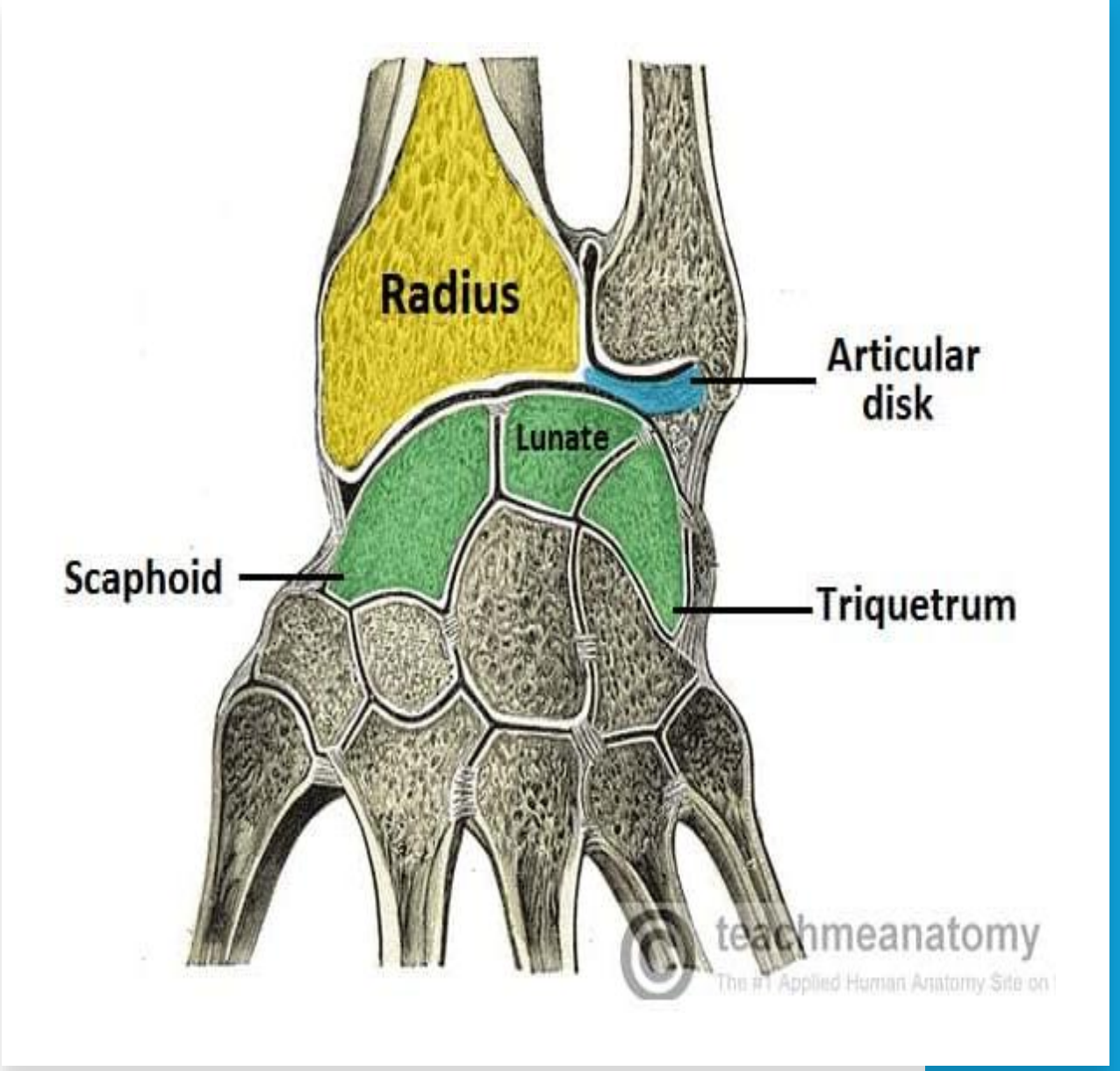




Wrist Joint

- The wrist joint (also known as the radiocarpal joint) is an articulation between the radius and the carpal bones of the hand.

- It is condyloid-type synovial joint which marks the area of transition between the forearm and the hand.



Capsule and Ligaments

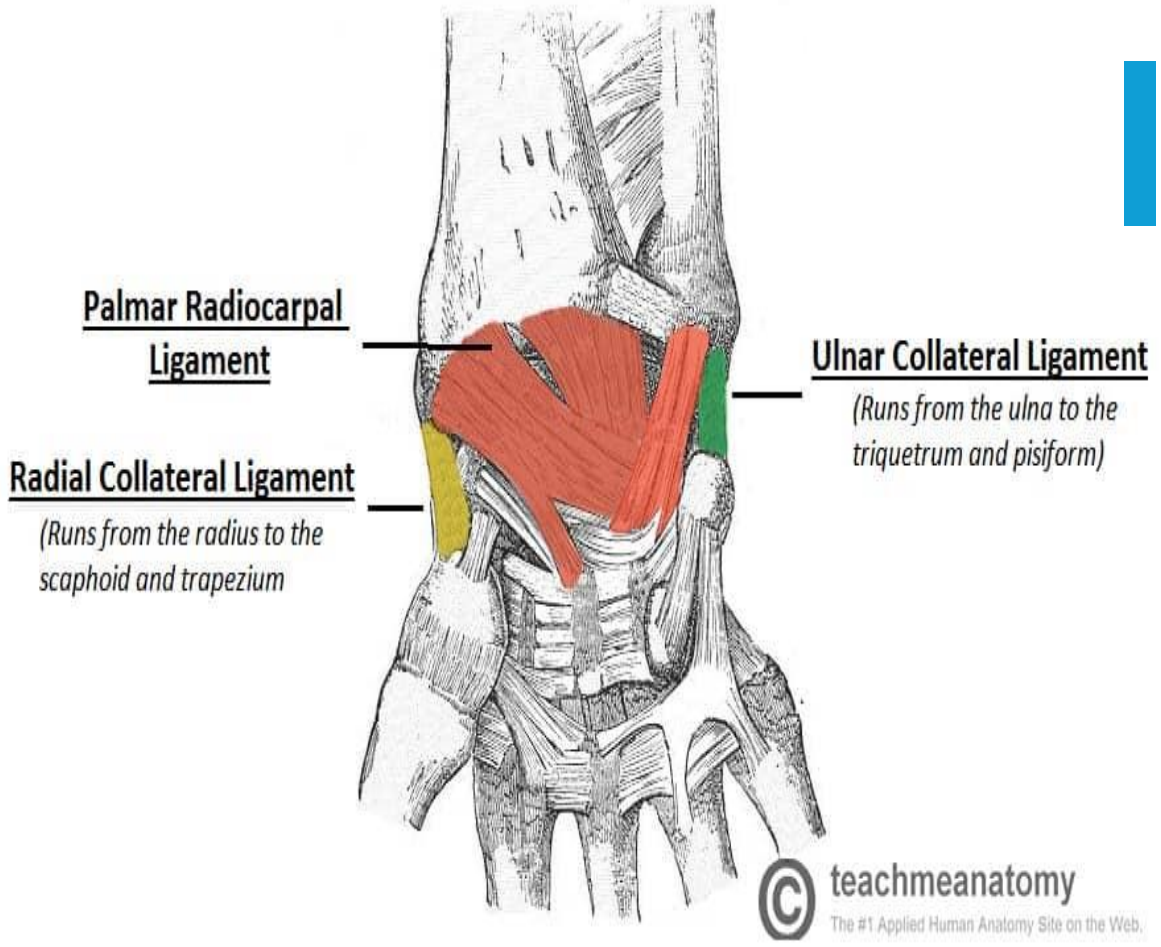
Joint Capsule

- The joint capsule of the wrist joint attaches to the radius, ulna and the proximal row of the carpal bones. It is lined internally by a synovial membrane, which produces synovial fluid to reduce friction between the articulating structures.

Ligaments

There are four main ligaments located at the wrist joint:

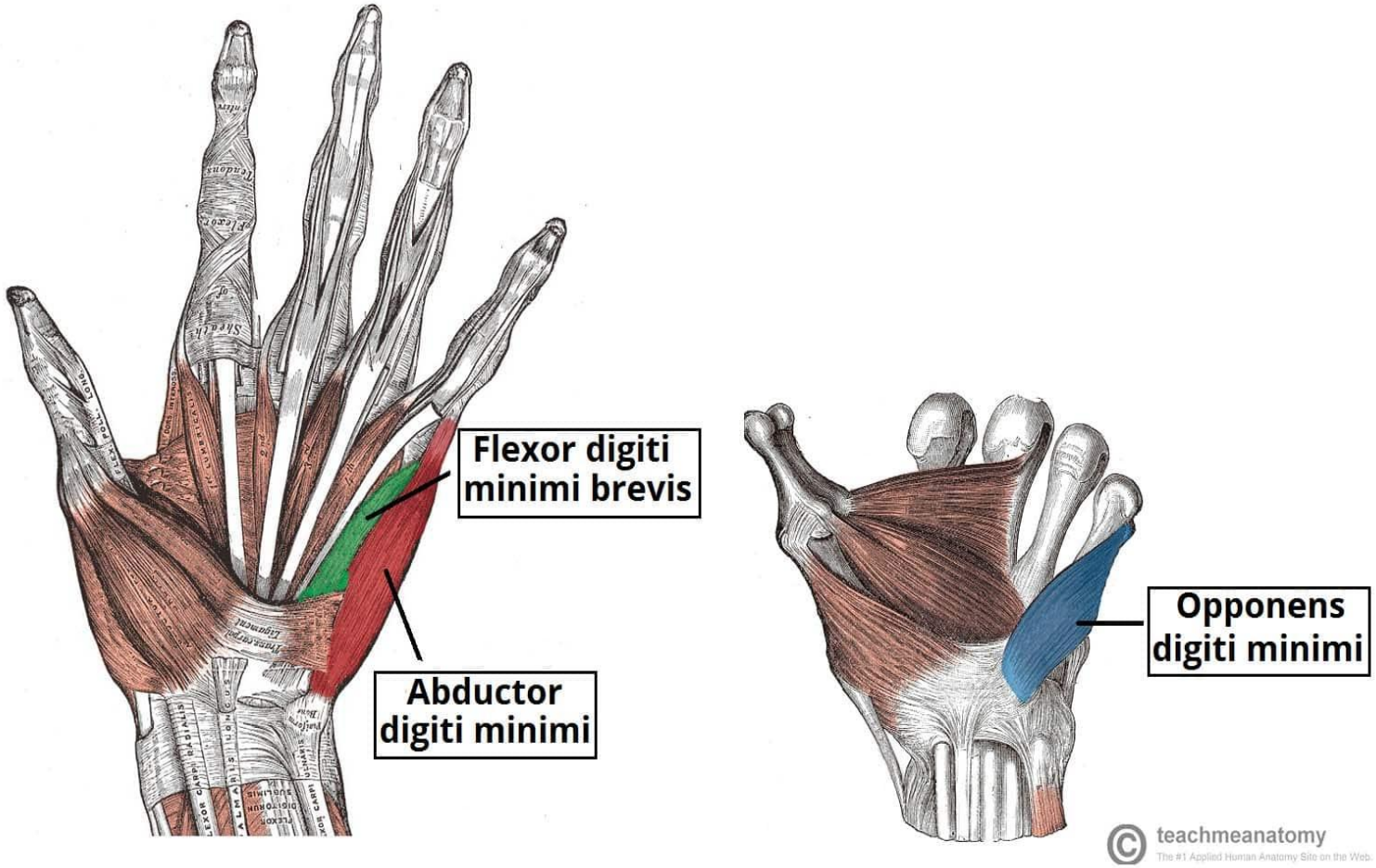
- Palmar radiocarpal
- Dorsal radiocarpal
- Ulnar collateral
- Radial collateral



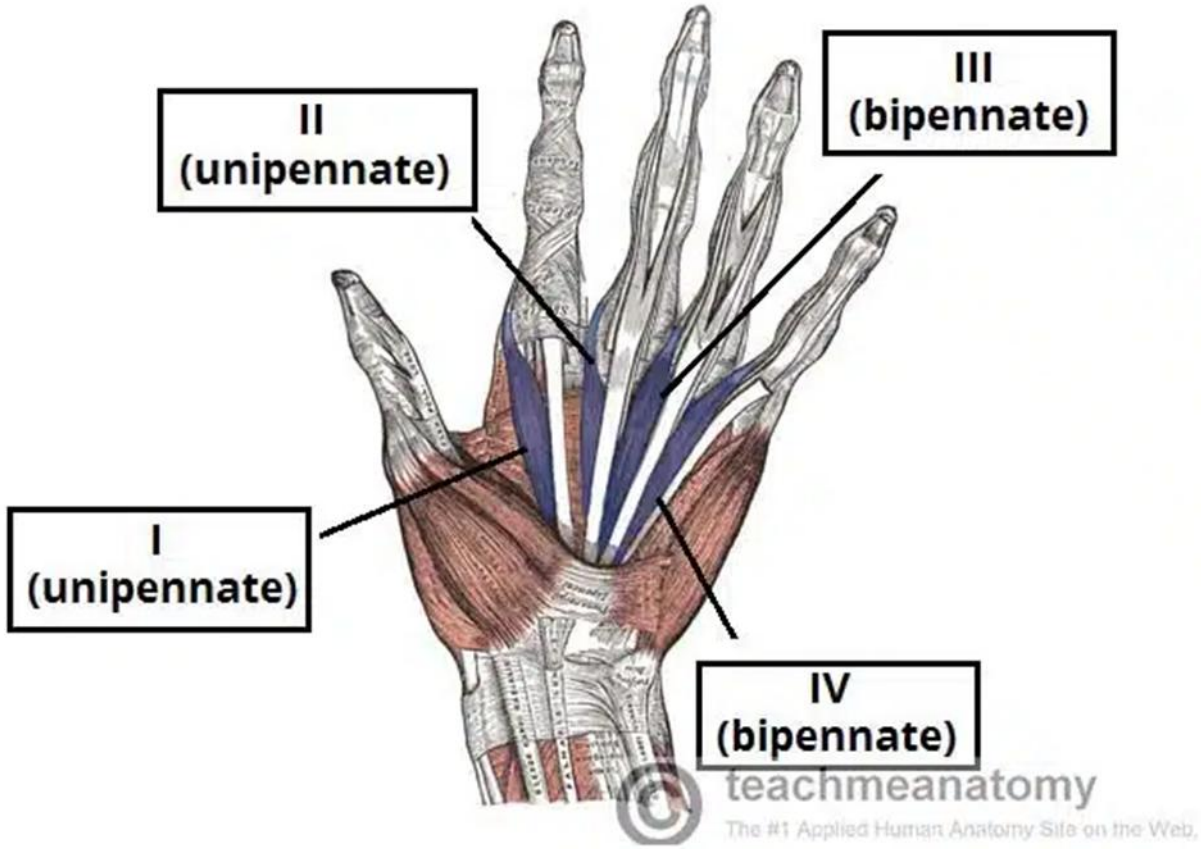
Hand Muscles

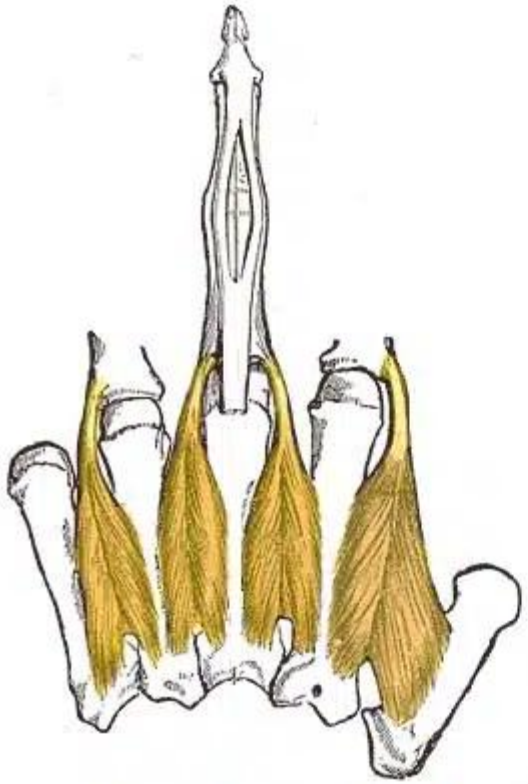
- The muscles that act on the hand can be divided into two groups:
    - **Extrinsic muscles** – located in the forearm. They control crude movements and produce a forceful grip.
    - **Intrinsic muscles** – located within the hand itself. They are responsible for the fine motor functions of the hand.
    - Thenar Muscles
    - Hypothenar Muscles
    - Lumbricals
    - Interossei
    - Other Muscles in the Palm

Thenar and Hypothenar Muscles

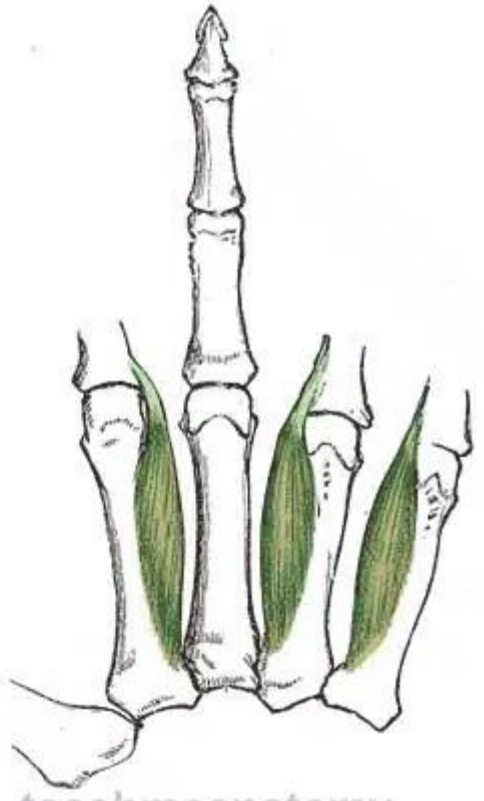


Lumbricals and interossei



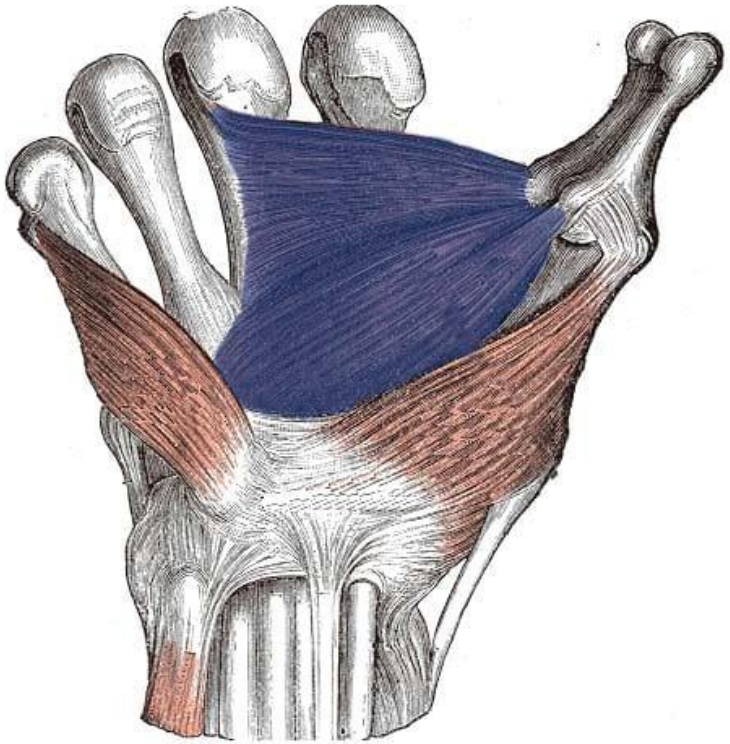


a) Dorsal Interossei



b) Palmar Interossei

Other Muscles in the Palm



- In your group explore the articulations in the wrist and hand and familiarise yourselves with the origin, insertion, nerve supply and action.

Muscles Mnemonics

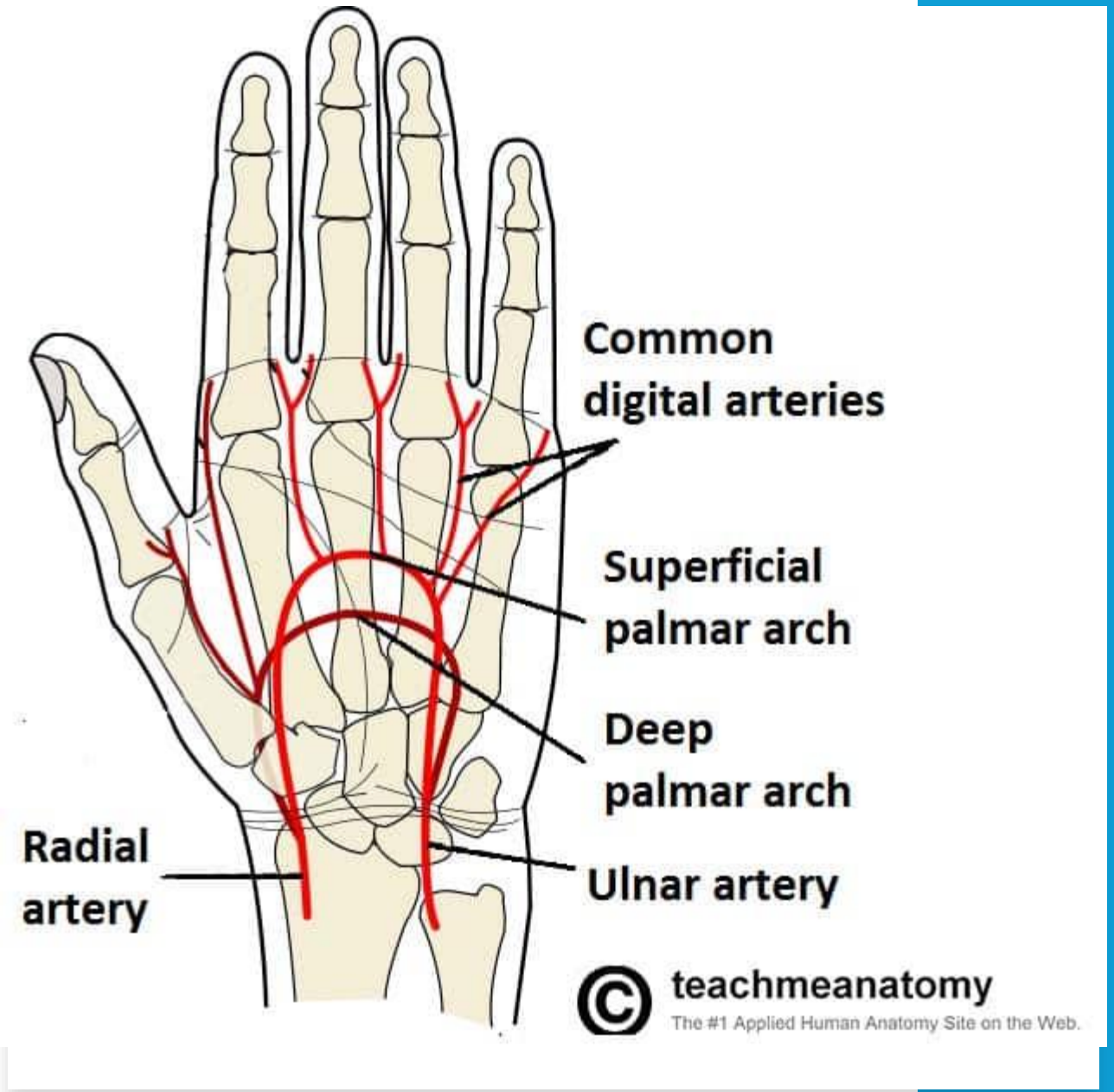
- Remembering the action of each muscles in the forerm can be quite difficult. Use the following mnemonic to make your life a little easier!

'Rule of 3s'

- **3 wrist flexors** (flexor carpi radialis, flexor carpi ulnaris, palmaris longus)
- **3 finger flexors** (flexor digitorum superficialis, flexor digitorum profundus, flexor pollicis longus)
- **3 wrist extensors** (extensor carpi radialis longus, extensor carpi radialis brevis, extensor carpi ulnaris)
- **3 finger extensors** (extensor digitorum, extensor indicis, extensor digiti minimi)
- **3 thumb extensors** (abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus)

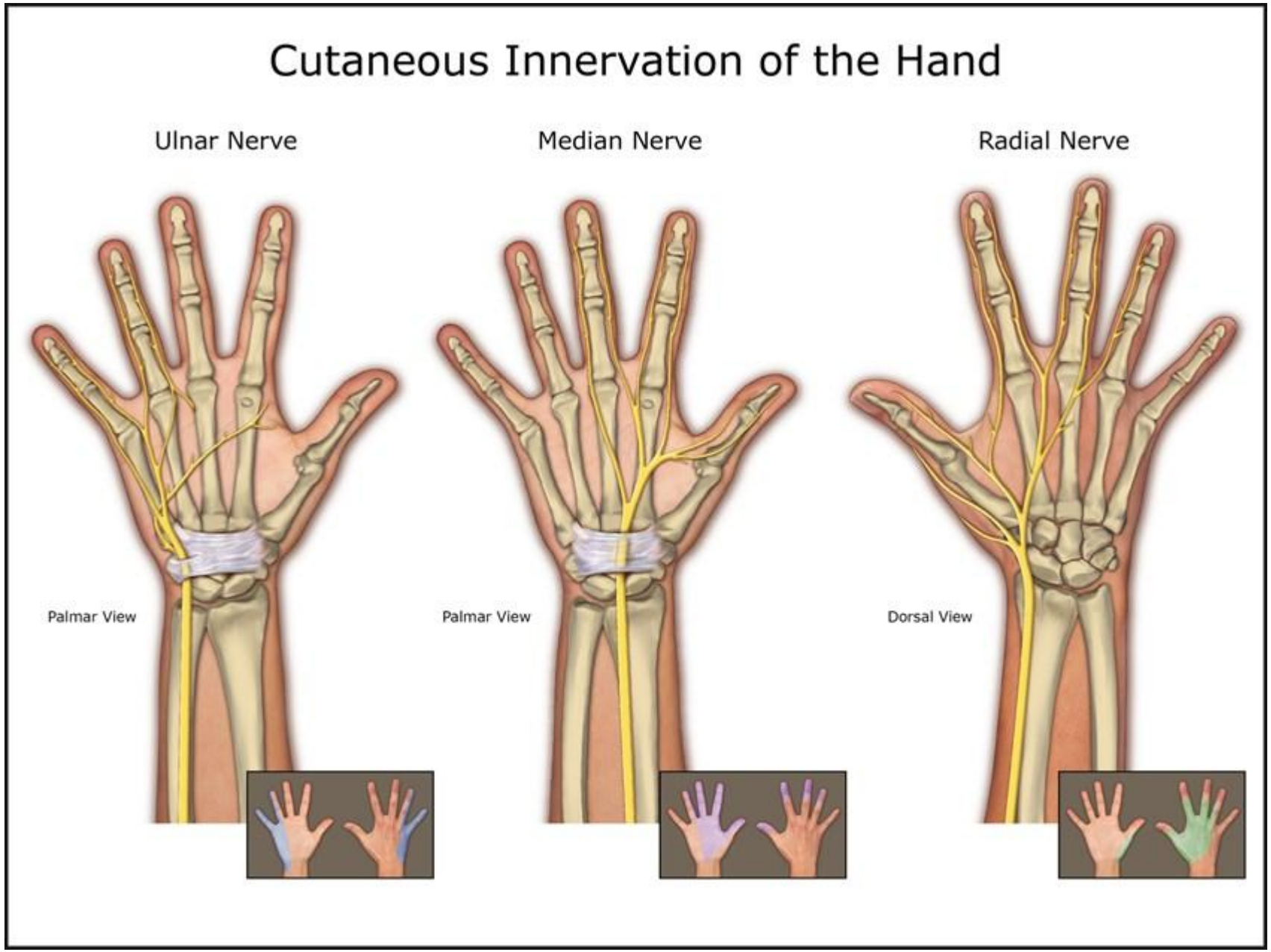
Circulation of the hand

- The wrist is an **ellipsoidal** (condyloid) type synovial joint, allowing for movement along two axes.
- This means that flexion, extension, adduction and abduction can all occur at the wrist joint.



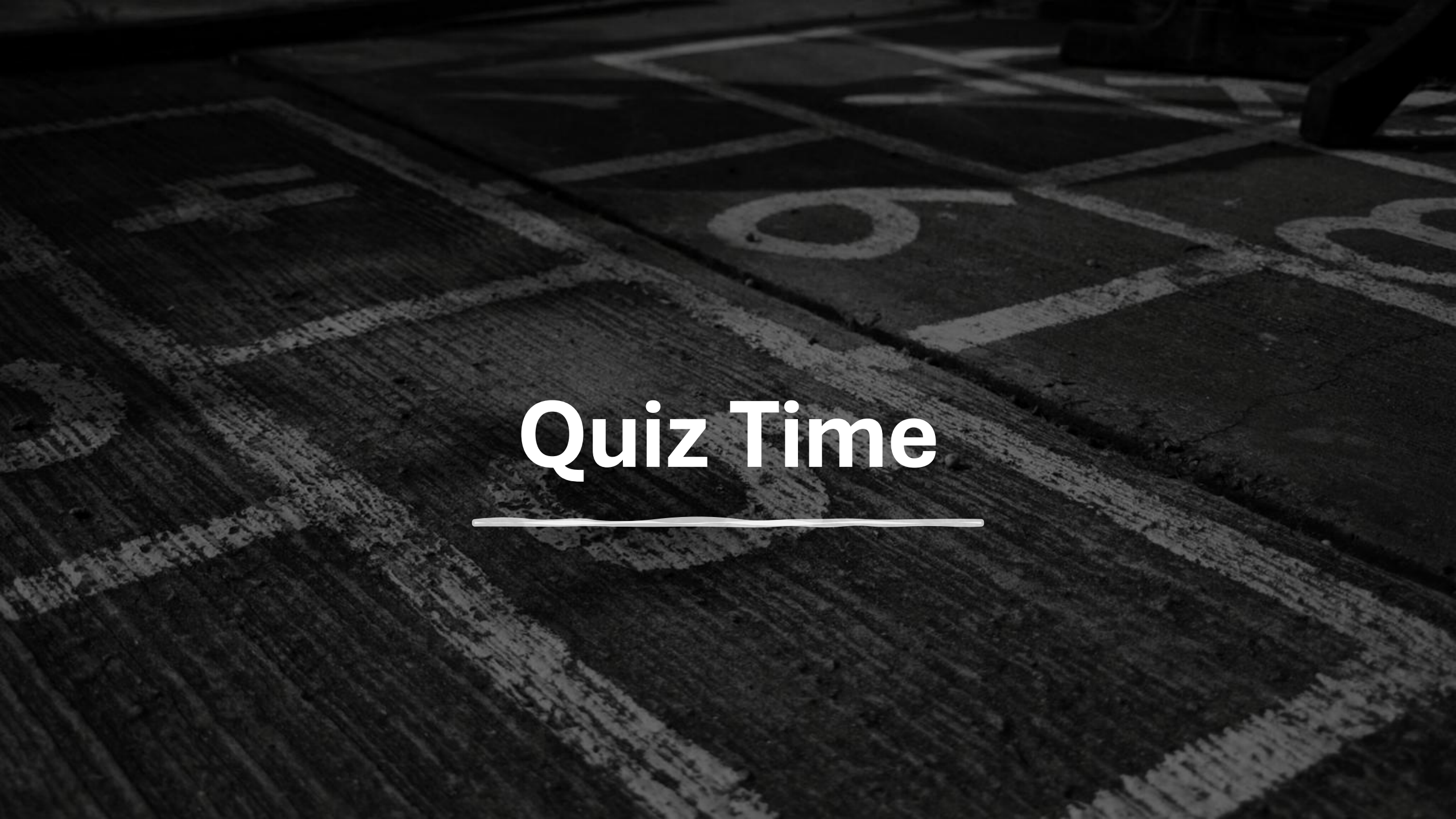
Innervation to the wrist is delivered by branches of three nerves:

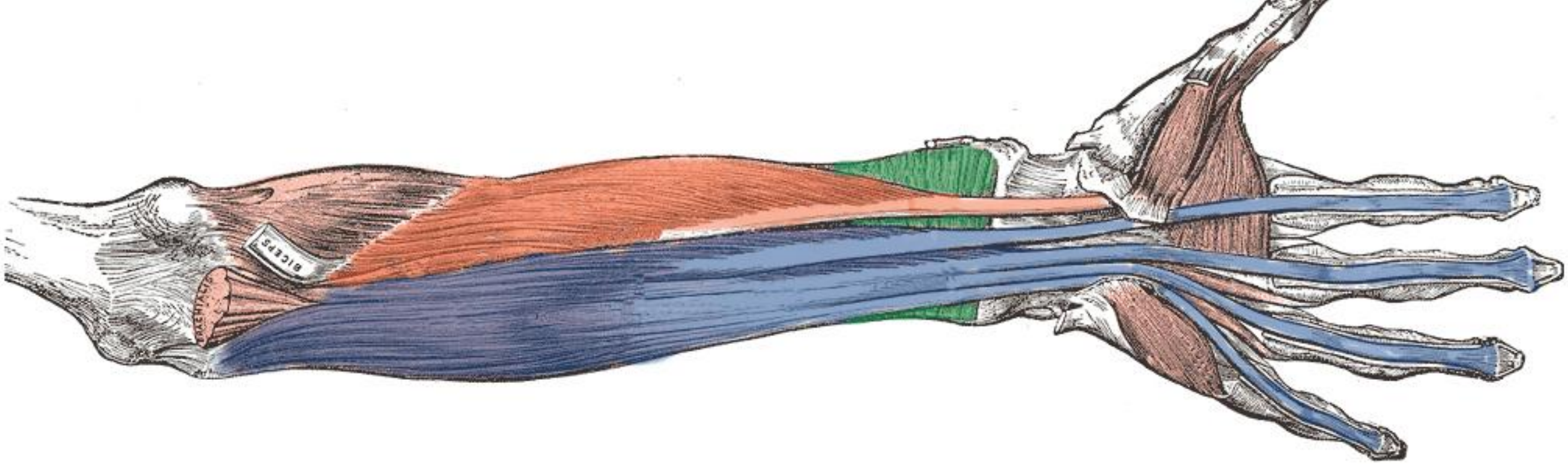
- Median nerve – Anterior interosseous branch.
- Radial nerve – Posterior interosseous branch.
- Ulnar nerve – deep and dorsal branches.



In your groups:

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Quiz Time



The illustration below demonstrates the anterior forearm muscles. Which muscle is demonstrated in green?

- Supinator
- Flexor policis longus
- Flexor digitorum profundus
- Pronator quadratus

Quiz

Where do the superficial muscles of the anterior forearm originate?

- Olecranon of the ulna
- Medial epicondyle of the humerus
- Lateral epicondyle of the humerus
- Deltoid tuberosity

Which of the following nerves, if damaged, would result in weakness of the flexor carpi radialis?

- Ulnar nerve
- Median nerve
- Radial nerve
- Musculocutaneous nerve

What is the action of brachioradialis?

- Elbow extension
- Elbow flexion
- Shoulder flexion
- Wrist pronation

What is the action of the lumbricals on the interphalangeal joints?

- Adduction
- Abduction
- Flexion
- Extension

Which nerve provides somatic motor innervation to the muscles of the posterior forearm?

- Median nerve
- Posterior cutaneous nerve of forearm
- Musculocutaneous nerve
- Radial nerve

Which bony landmark represents the common point of origin for the majority of muscles in the posterior compartment of the forearm?

- Lateral epicondyle
- Medial epicondyle
- Olecranon
- Deltoid tubercle

Which of the following muscles is considered an intrinsic muscle of the hand?

- Palmaris longus
- Extensor digiti minimi
- Adductor pollicis
- Extensor pollicis brevis

Which nerve innervates the opponens digiti minimi muscle of the hand?

- Median nerve
- Radial nerve
- Axillary nerve
- Ulnar nerve

Among the listed carpal bones, which one **does not** contribute to the articulatory surface of the wrist joint?

Scaphoid

Lunate

Triquetrum

Trapezoid

The wrist joint is formed by an articulation between the radius and proximal carpal bones. What type of synovial joint is it?

Pivot joint

Saddle joint

Hinge joint

Ellipsoid joint

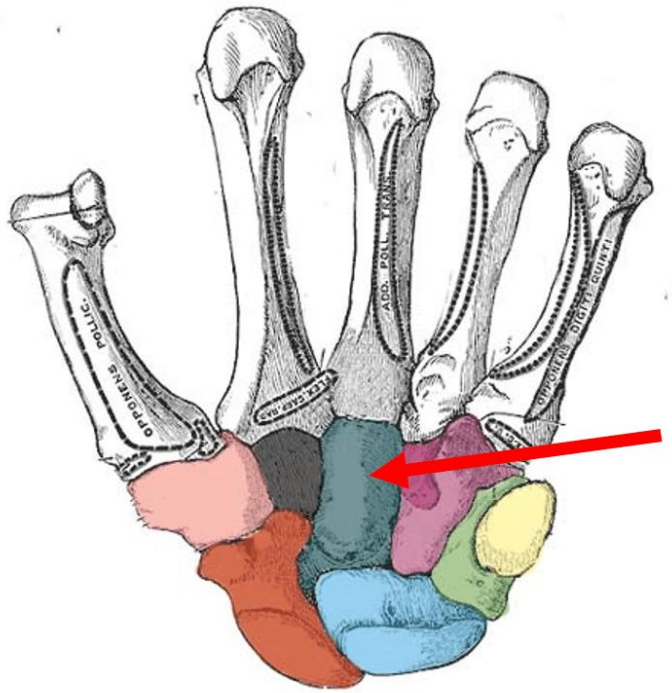
Below is an illustration of the carpal bones of the hand. Which bone is denoted by the arrow?

Trapezoid

Lunate

Hamate

Capitate

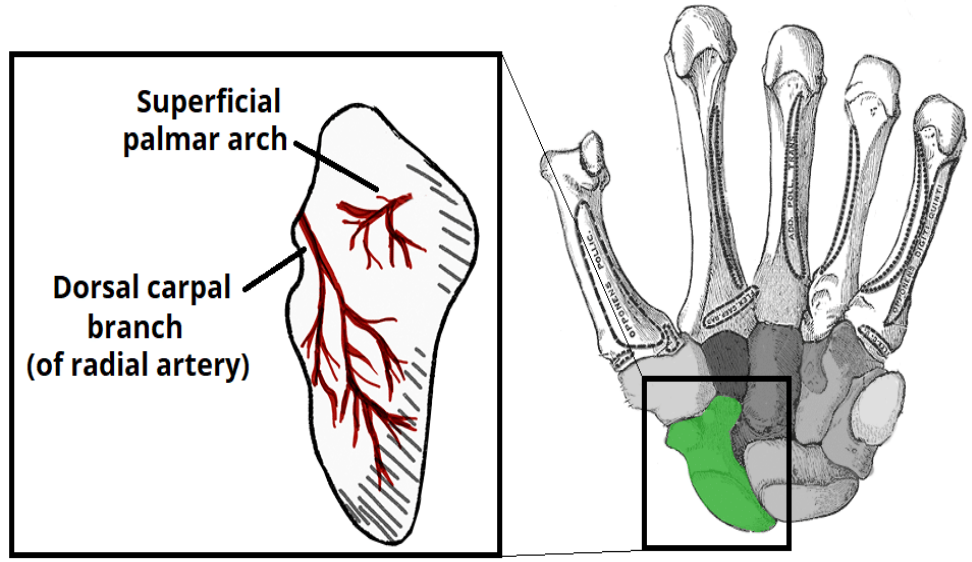




Some of the injuries of the forearm and hand

Scaphoid Fracture

- The scaphoid bone of the hand is the most commonly fractured carpal bone – typically by falling on an oustretched hand (FOOSH).
- In a fracture of the scaphoid, the characteristic clinical feature is pain and tenderness in the anatomical snuffbox.
- The scaphoid is at particular risk of avascular necrosis after fracture.
- Patients with a missed scaphoid fracture are likely to develop osteoarthritis of the wrist in later life.





Anterior Dislocation of the Lunate

- This can occur by falling on a dorsiflexed wrist. The lunate is forced anteriorly, and compresses the carpal tunnel, causing the symptoms of carpal tunnel syndrome.

- This manifests clinically as paraesthesia in the sensory distribution of the median nerve and weakness of thenar muscles. The lunate can also undergo avascular necrosis, so immediate clinical attention to the fracture is needed.

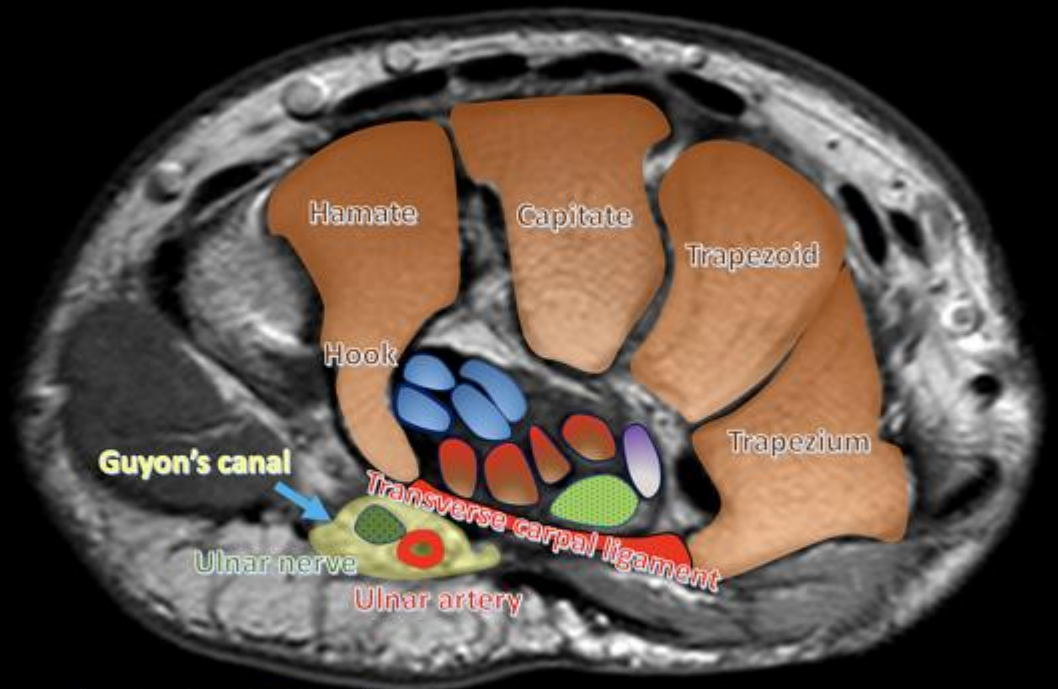


Colles’ Fracture

- The Colles’ fracture is the most common fracture involving the wrist, caused by falling onto an outstretched hand.
- The radius fractures, with the distal fragment being displaced posteriorly. The ulnar styloid process can also be damaged, and is avulsed in the majority of cases.
- This clinical condition produces what is known as the ‘dinner fork deformity’.

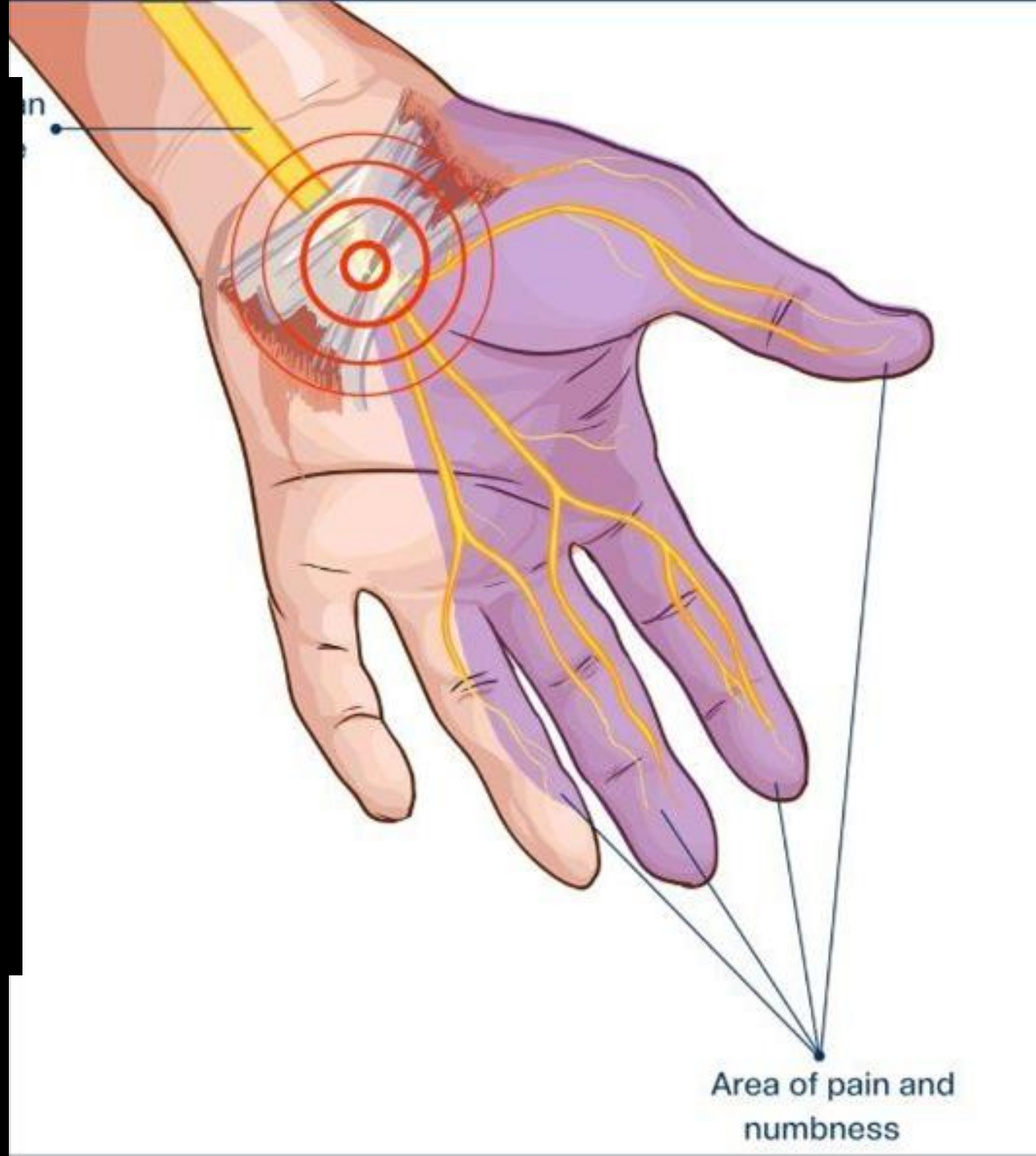


Carpal Tunnel Anatomy

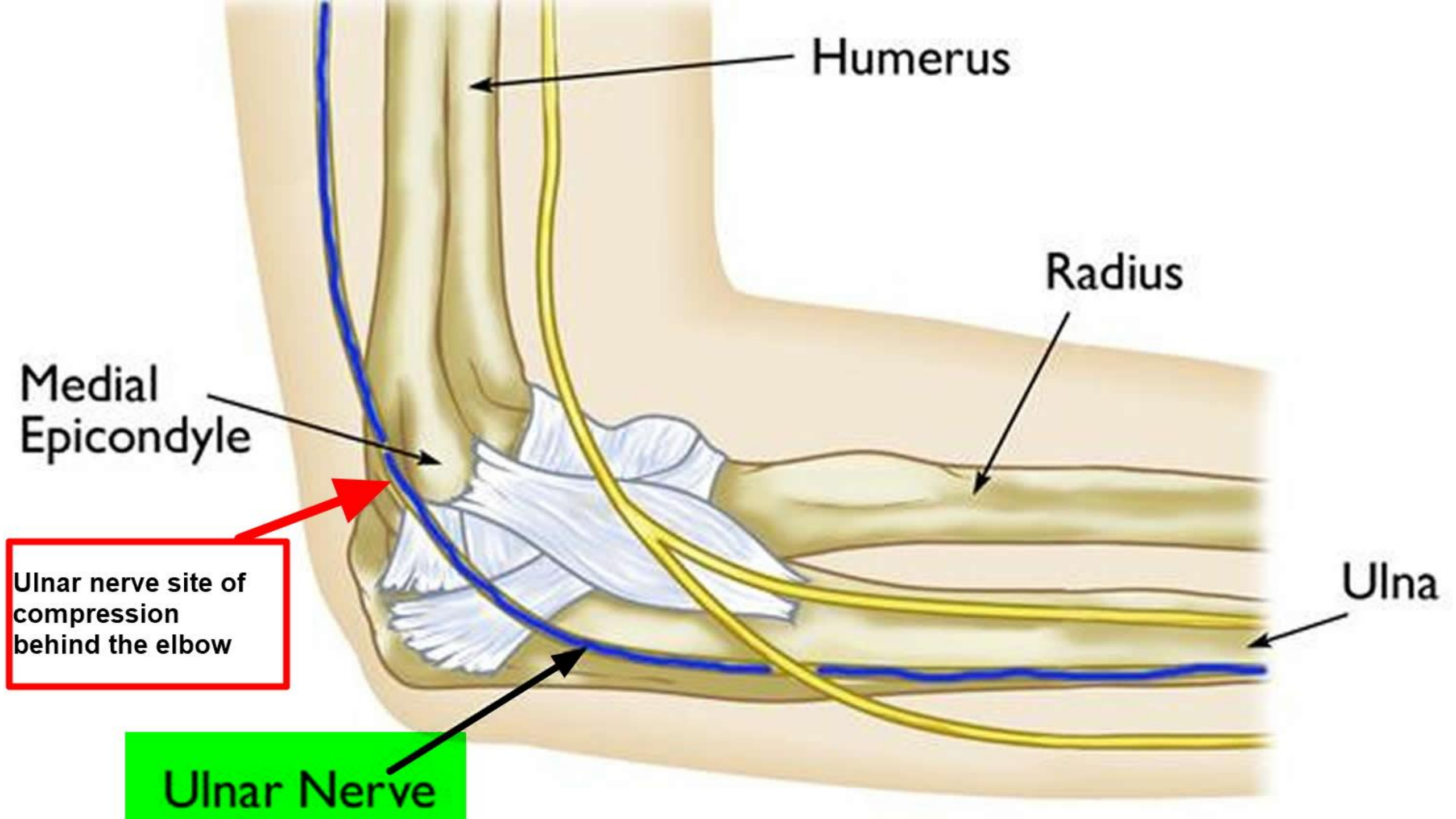


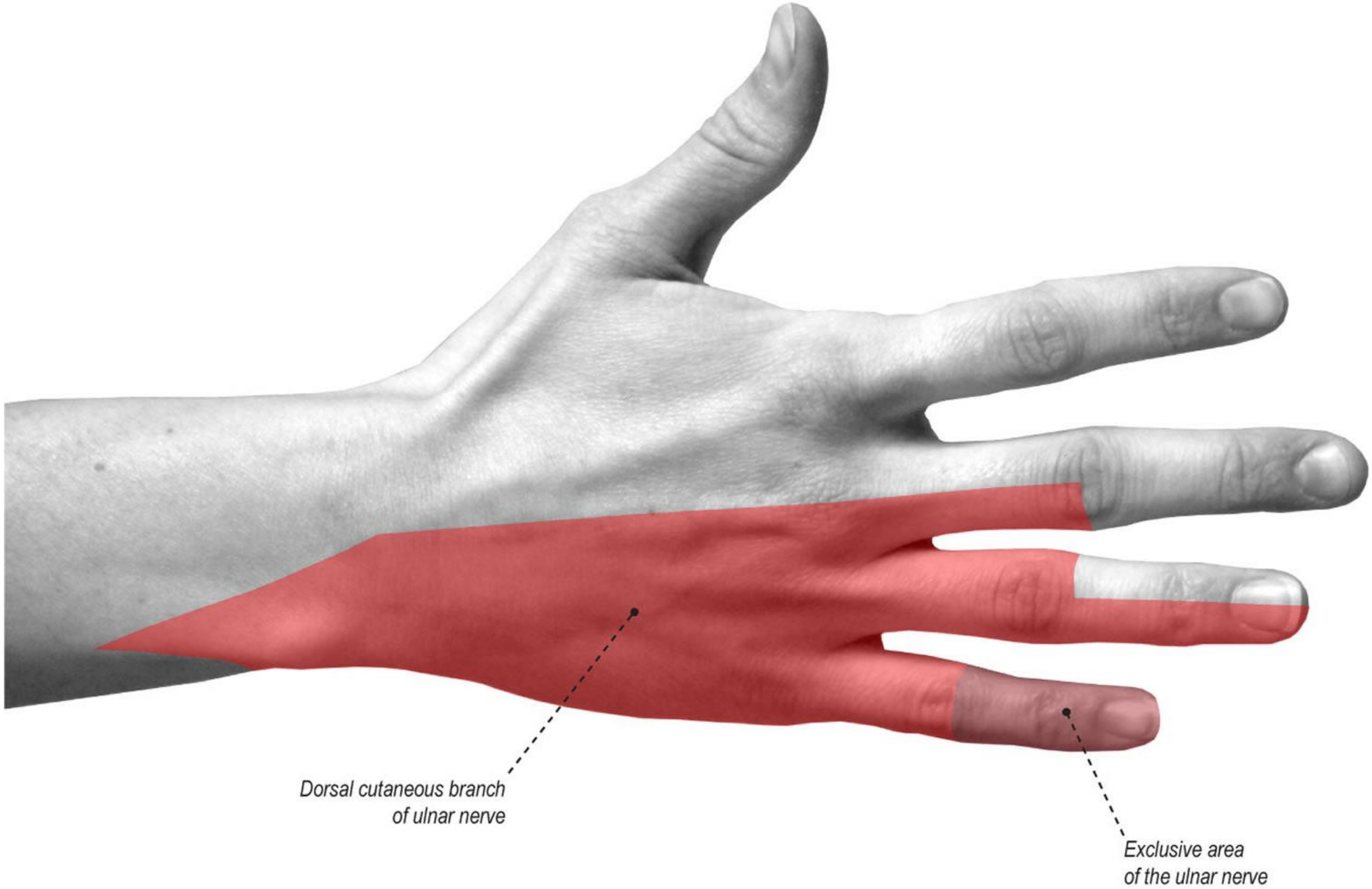
Carpal tunnel contents:

- Median nerve
- Flexor digitorum superficialis
- Flexor digitorum profundus
- Flexor pollicis longus



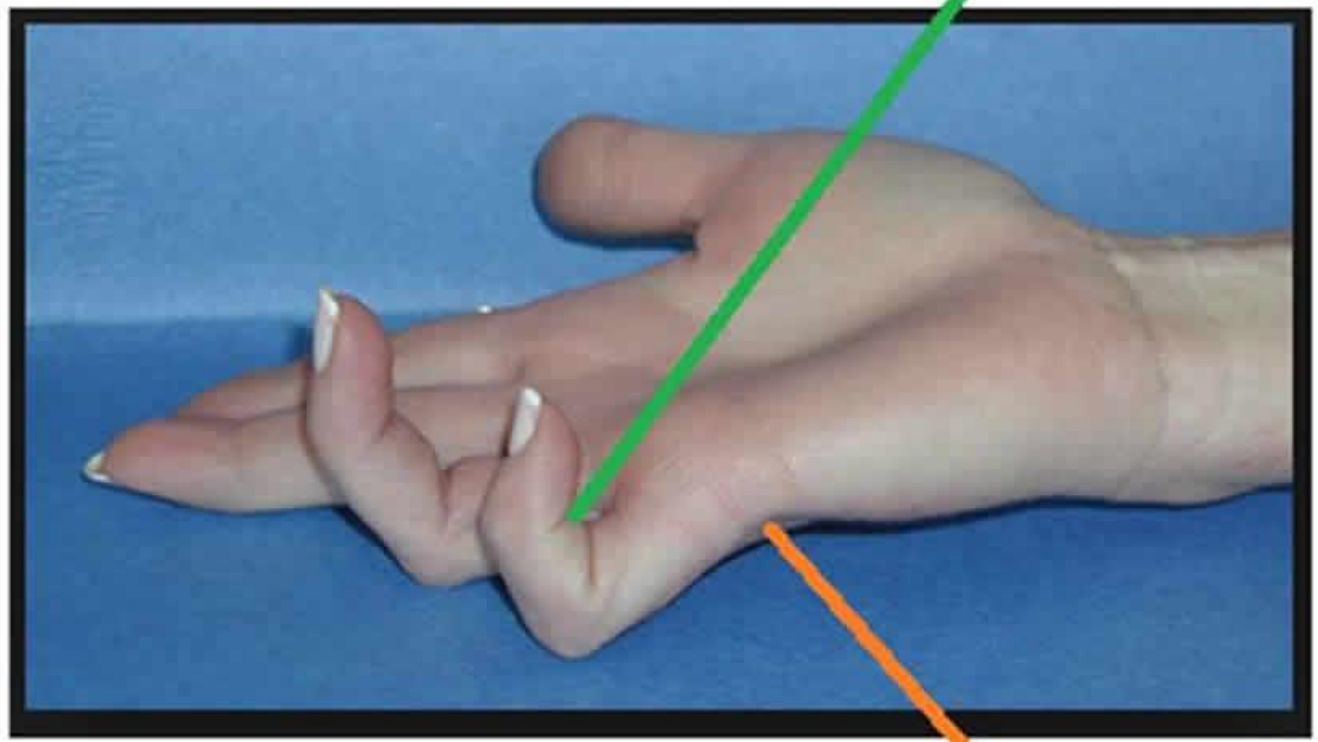
Carpal Tunnel Syndrome (CTS)


Humerus
Radius
Medial Epicondyle
Ulnar nerve site of compression behind the elbow
Ulna
Ulnar Nerve


Dorsal cutaneous branch
of ulnar nerve
Exclusive area
of the ulnar nerve

Ulnar Claw Hand (Paralysed lumbricals for ring & little fingers)

Interphalangeal flexion (unopposed flexor digitorum muscles)



Hyperextension at metacarpophalangeal joint (unopposed extensor digitorum muscles)

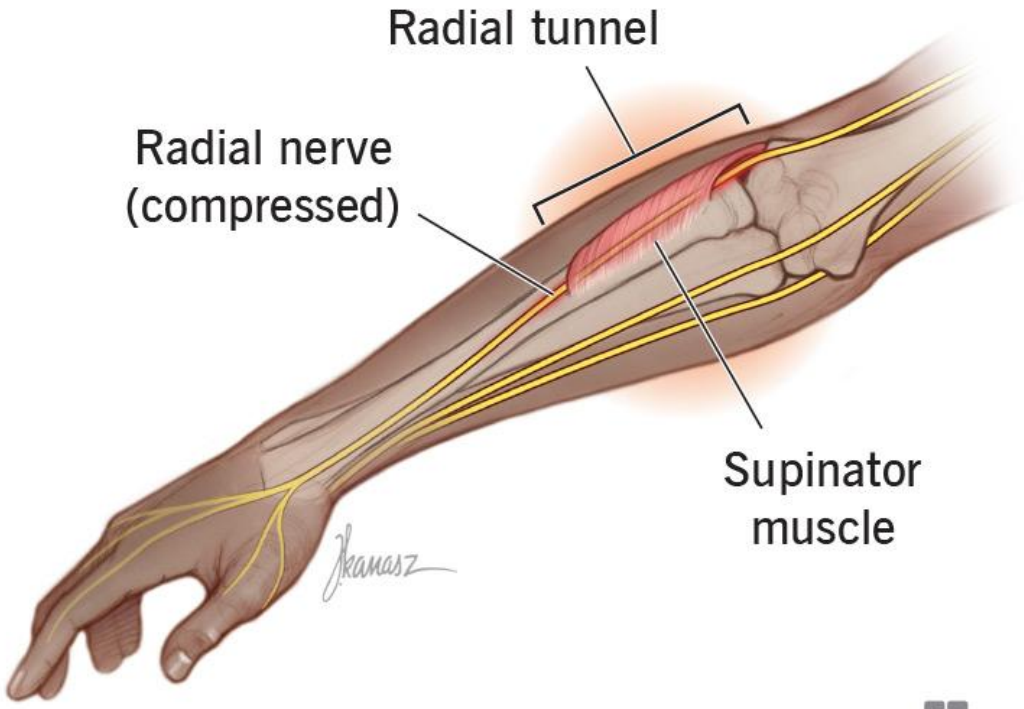
Wrist drop is a sign of radial nerve injury that has occurred proximal to the elbow.

There are two areas where the radial nerve is at high-risk of damage:

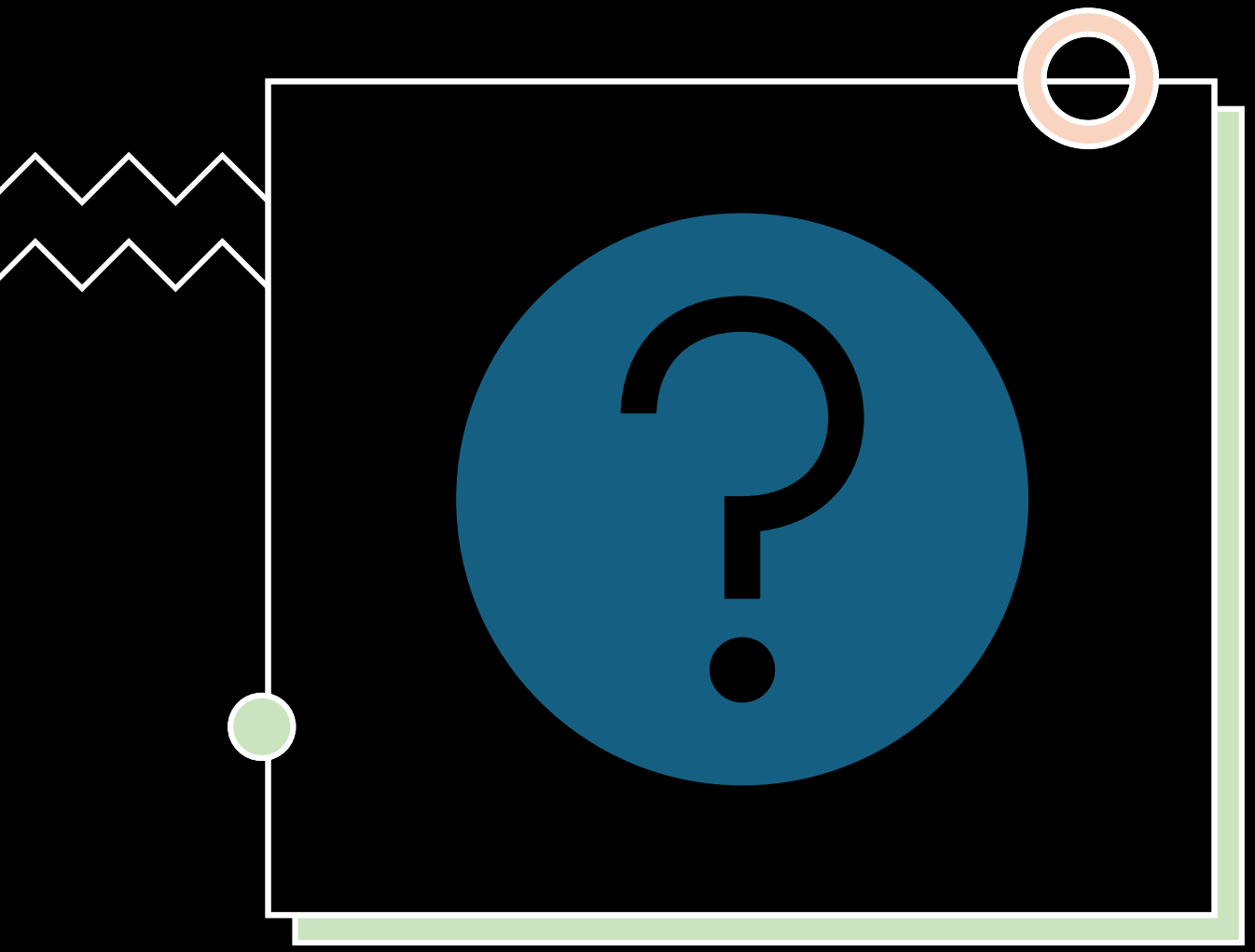
- **Axilla** – can be injured via humeral dislocations or fractures of the proximal humerus.
- **Radial groove of the humerus** – can be injured via a humeral shaft fracture.
- The radial nerve innervates all muscles in the **extensor compartment of the forearm**. In the event of a radial nerve lesion, these muscles are paralysed. The muscles that flex the wrist are innervated by the median nerve, and thus are unaffected. The tone of the flexor muscles (and gravity) produces unopposed flexion at the wrist joint.



Radial tunnel syndrome



Cleveland
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Questions